# Mathematical Logic

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## 1 Introduction

*Logic focuses on how to reliably get from assumptions to conclusions.*

Whether or not $F = ma$ is true is not the concern of the logician. What they ask instead is *if $F = ma$, does it follow that $a = \frac{F}{m}$?* This question isn't limited to the given formula alone, i.e., it generalizes. Although logic may seem common sense at first glance, our common sense often run into problems (paradoxes).

**Barber Paradox:** *In a town there is one barber. This barber shaves people if and only if they don't shave themselves. The question is, does the barber shave himself?*

The above is a simpler stating of Russell's famous paradox.

**Russell's Paradox:** Let $R = \{X : X \text{ is a set and } X \notin X\}$. The question is, is $R \in R$? If so, the $R$ belongs to a set that don't contain themselves. Therefore, $R \notin R$. And if $R \notin R$, then $R$ is a set that doesn't contain itself, thus $R \in R$. Hence, the paradox.

**Berry's Paradox:** Let $B$ be the set of all natural numbers that can be described by an english sentence of at most 200 characters. Is $B$ finite? It seems so since there are at most $26^{200}$ combinations of characters and each combination describes at most one number. Hence, $B$ should be finite. Consider *the first number that can not be described by an English sentence of at most 200 characters.* Since $B$ is finite, such a number exists. But the description we wrote for it is less than 200 characters.

These paradoxes demonstrate the need for a formal, rigorous system of logic. So our goal is to develop tools that'll help us move from assumption to conclusions.

As seen earlier, our natural languages are not fit for this task. We need to construct and analyze formal languages. We will focus on two such languages: Propositional Logic (PL) and First Order Logic (FOL). Other types of logic, such as temporal logic, will also be introduced.

In constructing languages, we must distinguish between two components: **Syntax** and **Semantics**. In short, syntax is concerned with structure (form) and semantics is concerned with entailment (meaning).

When defining the language of PL , we need to specify an infinite set of valid strings, commonly called formulas. Because the set is infinite and its members do not share a single easily expressible common property, we cannot define it directly using set-builder notation. Instead, we rely on an inductive definition.

An inductive definition of a set consists of:

1. A universe set $X$.
2. A core (base) set $B \subseteq X$.
3. A finite set of operations, i.e., functions $\mathcal{F} = \bigcup_{n \in \mathbb{N}} \mathcal{F}_n$ where $\mathcal{F}_n$ consists of functions $f : A^n \to A$.
4. The concept of closure, i.e., $A \subseteq X$ is closed if for any function $f$ with arity $k$ and elements $a_1, \dots, a_k \in A$, $f(a_1, \dots, a_k) \in A$.

A set defined inductively from $(X, B, \mathcal{F})$ is denoted $I(X, B, \mathcal{F})$ or simply $I$ when the context is clear. It is defined as the smallest subset of $X$ that contains $B$ and is closed under $\mathcal{F}$. Let $\mathcal{C} = \{A : B \subseteq A \text{ and } A \text{ is closed under} \mathcal{F}\}$ Let $I$ be defined as follows:

$$I = \bigcap \mathcal{C}$$

**Proposition 1.1.** *The following hold*

*i)* $B \subseteq \bigcap \mathcal{C}$

*ii)* $\bigcap \mathcal{C}$ *is closed under* $\mathcal{F}$

*iii)* $\bigcap \mathcal{C}$ *is the minimal set satisfying (i) and (ii).*

*iv) No other set can satisfy (i), (ii), and (iii) simultaneously.*

*Proof.* i) Trivial.

ii) Let $x \in \bigcap \mathcal{C}$ then $x \in A$ for all $A$ closed under $\mathcal{F}$. Therefore, $f(a) \in A$ for all $A$ and for all $f \in \mathcal{F}$. Hence, $f(a) \in \bigcap \mathcal{C}$.

iii) Let $\mathcal{D}$ be another set satisfying (i) and (ii). Then $\mathcal{D} \in \mathcal{C}$. Thus, $\bigcap \mathcal{C} \subseteq \mathcal{D}$.

iv) Let $\mathcal{E}$ and $\bigcap \mathcal{C}$ be two sets that satisfy (i), (ii), and (iii). By minimality, we have $\mathcal{E} \subseteq \bigcap \mathcal{C}$ and $\bigcap \mathcal{C} \subseteq \mathcal{E}$. Hence, $\mathcal{E} = \bigcap \mathcal{C}$. □

# 2 The Language of Propositional Logic (PL)

We define our language as follows. We first fix a set of symbols. These symbols are of two kinds:

**1. Logical Symbols:**

$$), (, \neg, \vee, \wedge, \rightarrow$$

**2. Non-logical (Variable) Symbols:**

$$p_1, p_2, p_3, \dots$$

Next, we build our set of expressions. An expression is a finite sequence of symbols. Some expressions can be pure nonsense, such as:

$$((p_1 \rightarrow$$

We want to define propositions or well-formed formulas (wffs) as "grammatically correct" expressions; the nonsensical expressions must be excluded. We define them inductively as follows:

a. Every variable symbol is a proposition (wff).

b. If $\alpha$ and $\beta$ are wffs, then so are $\neg\alpha, (\alpha \vee \beta), (\alpha \wedge \beta)$, and $(\alpha \to \beta)$.

c. No expression is a wff unless it is compelled to be one by (a) and (b).

To determine whether a given expression is a well-formed formula (wff) in the language of PL, we check whether it can be constructed inductively from the atomic formulas (propositional variables) using the formation rules. An ancestral tree (or formation tree) visualizes this inductive construction: the root is the full formula, internal nodes correspond to applications of connectives, and the leaves are atomic propositions. Consider the expression

$$\neg(((p_1 \vee p_2) \to p_3) \wedge (p_3 \vee \neg p_4)).$$

The following ancestral tree shows that it is indeed a well-formed formula, as it can be built step by step from the atomic formulas $p_1, p_2, p_3, p_4$:

$$\neg(((p_1 \vee p_2) \to p_3) \wedge (p_3 \vee \neg p_4))$$

$$((p_1 \vee p_2) \to p_3) \qquad\qquad (p_3 \vee \neg p_4)$$

$$(p_1 \vee p_2) \qquad p_3 \qquad\qquad p_3 \qquad \neg p_4$$

$$p_1 \qquad p_2 \qquad\qquad\qquad\qquad p_4$$

Hence, our expression is a well-formed formula.

**Theorem 2.1.** *$\alpha$ is a well-formed formula if and only if $\alpha$ has a valid ancestral tree.*

*Proof.* ($\Rightarrow$) If $\alpha$ is a well-formed formula, then by the inductive definition of the set of wffs, it can be constructed from atomic formulas by finitely many applications of the formation rules. This construction corresponds exactly to a valid ancestral tree with root $\alpha$.

($\Leftarrow$) If $\alpha$ has a valid ancestral tree, then the tree provides a finite inductive construction of $\alpha$ from atomic formulas using the formation rules. Hence $\alpha$ belongs to the smallest set containing the atomic formulas and closed under the formation rules, i.e., $\alpha$ is a well-formed formula. $\square$

**Exercise 2.0**

Show that there are no wffs of length 2, 3, or 6, but that any other positive length is possible.

*Proof.* We proceed with structural induction.
**Base Case:** Let $\alpha$ be a propositional variable, then $|\alpha| = 1$. Hence, the claim holds.
**Inductive Step:** Let $\alpha$ and $\beta$ satisfy the claim. The $|(\neg\alpha)| = 3 + |\alpha|$. Cases for $|(\neg\alpha)| = 2$ and $|(\neg\alpha)| = 3$ clearly fail. If $|(\neg\alpha)| = 6$, then $|\alpha| = 3$, contradicting our inductive hypothesis. Thus our claim holds for $(\neg\alpha)$.
Next, we have $|(\alpha \vee \beta)| = 3 + |\alpha| + |\beta|$. Then cases for $|(\alpha \vee \beta)| = 2$ and $|(\alpha \vee \beta)| = 3$ clearly fail. If $|(\alpha \vee \beta)| = 6$, then $|\alpha| + |\beta| = 3$. Which means one of $\alpha$ or $\beta$ has length 2, contradicting our inductive hypothesis. Hence our claim holds for $|(\alpha \vee \beta)|$. The proofs for the other connectives are analogous. $\square$

## Exercise 2.1

Let $\alpha$ be a wff; let $c$ be the number of places at which binary connective symbols $(\wedge, \vee, \rightarrow, \leftrightarrow)$ occur in $\alpha$; let $s$ be the number of places at which sentence symbols occur in $\alpha$. (For example, if $\alpha$ is $(\mathbf{A} \rightarrow (\neg\mathbf{A}))$ then $c = 1$ and $s = 2$.) Show by using the induction principle that $s = c + 1$.

*Proof.* Again, we proceed by structural induction.
**Base Case:** Let $\alpha$ be a propositional variable. Then $c = 0$ and $s = 1$. Hence $s = c + 1$ holds.
**Inductive Step:** Let $\alpha$ and $\beta$ have our desired property. Then for $(\neg\alpha)$ nothing changes. By our inductive hypothesis $s_{(\neg\alpha)} = c_{(\neg\alpha)} + 1$. For $(\alpha \vee \beta)$, note that $c_{(\alpha\vee\beta)} = c_\alpha + c_\beta + 1$. And we have $s_{(\alpha\vee\beta)} = s_\alpha + s_\beta = (c_\alpha + 1) + (c_\beta + 1) = (c_\alpha + c_\beta + 1) + 1$. Hence the property holds. The cases for the other connectives are analogous. □

## Exercise 2.2

Assume we have a construction sequence ending in $\varphi$, where $\varphi$ does not contain the symbol $\mathbf{A}_4$. Suppose we delete all the expressions in the construction sequence that contain $\mathbf{A}_4$. Show that the result is still a legal construction sequence.

*Proof.* Let $S = \{\alpha_1, \dots, \alpha_n\}$ be a construction sequence such that $\alpha_n = \varphi$. Let $S'$ be the sequence obtained by removing all $\alpha_i$ containing $\mathbf{A}_4$. Clearly, $\varphi \in S'$.
We now check that $S'$ is still a legal construction sequence. Take any $\beta \in S'$.

- If $\beta$ is a variable, it is trivially allowed.
- If $\beta = \neg\gamma$, then $\gamma$ does not contain $\mathbf{A}_4$ (otherwise $\beta$ would contain $\mathbf{A}_4$) and hence $\gamma \in S'$ appearing before $\beta$.
- If $\beta = (\gamma_1 * \gamma_2)$ for $* \in \{\wedge, \vee, \rightarrow\}$, then $\gamma_1, \gamma_2$ do not contain $\mathbf{A}_4$ and hence appear in $S'$ before $\beta$.

Thus, all formulas in $S'$ satisfy the rules of a construction sequence. Hence $S'$ is a legal construction sequence ending in $\varphi$. □

## Exercise 2.3

Suppose that $\alpha$ is a wff not containing the negation symbol $\neg$.
(a) Show that the length of $\alpha$ (i.e., the number of symbols in the string) is odd.
(b) Show that more than a quarter of the symbols are sentence symbols.

*Proof.* (Based on the suggestion given on the textbook, we show that the length of $\alpha$ is $4k + 1$ and the number of sentence symbols is $k + 1$.) We proceed by induction.
**Base Case:** For a propositional variable $\alpha$, $|\alpha| = 1$, i.e., $4k + 1$ for $k = 0$. (b) holds as well since the number of sentence symbols is $1 = k + 1$.
**Inductive Step:** Let $\alpha$ and $\beta$ satisfy the desired property. Then for $(\alpha*\beta)$ where $* \in \{\wedge, \vee, \leftarrow\}$, $|(\alpha * \beta)| = 3 + |\alpha| + |\beta| = 3 + (4m + 1) + (4n + 1) = 4(m + n + 1) + 1$ for some $m, n \in \mathbb{N}^+$. As for the number of sentence symbols, see that $(\alpha * \beta) = (m + 1) + (n + 1) = (m + n + 1) + 1$. Set $k = m + n + 1$, then $|(\alpha * \beta)| = 4k + 1$ and the number of sentence symbols is $k + 1$. Hence, both (a) and (b) hold. □

# 3 Syntactic Properties of Propositional Formulas

*Given any statement $p$, can we be able to show that it doesn't belong to the set of well-formed formulas?*

A basic intuition that can help us solve this problem is: find a property (in this case a subset) $\mathsf{T}$ such that only the set of all well-formed formulas are contained in $\mathsf{T}$ but $p \notin \mathsf{T}$. Hence we need a tool for showing that some property $\mathsf{T}$ holds for all wffs.

Consider the following property: *every wff is either a propositional variable or starts with a left-bracket.* We prove this by induction.

*Proof.* **Base Case:** Every propositional variable is a wff, by definition.

**Inductive Step:** If $\alpha$ and $\beta$ are propositional variables, then clearly $(\neg\alpha), (\alpha \wedge \beta), (\alpha \vee \beta), (\alpha \rightarrow \beta)$ start with a left bracket. $\square$

Then our set $\mathsf{T}$ consists of those expressions that are either a propositional variable or start with a left bracket. Clearly, the set of well-formed formulas is contained in $\mathsf{T}$. From this we get a principle.

**The generalized induction principle:** *If a property satisfied by all members of $B$ and if it is closed under $\mathcal{F}$, then every member of $I(X, B, \mathcal{F})$ holds that property.*

We can now use this to prove multiple properties of the set of wffs. For example, we can show that in any wff, the number of left and right brackets are balanced (try the proof).

Next, we define the proper initial segment of a wff. Let $\varphi = s_1 s_2 \dots s_n$ be a well-formed formula (wff), viewed as a finite sequence of symbols. A *proper initial segment* of $\varphi$ is any sequence of the form

$$s_1 s_2 \dots s_k$$

where $1 \leqslant k < n$.

**Claim:** If $\alpha$ is a wff and $\gamma$ is an initial segment of $\alpha$, then the number of right brackets in $\gamma$ is strictly less that the number of left brackets in $\gamma$.

*Proof.* We use structural induction.

**Base Case:** Let $\alpha$ be an atomic formula, then our statement is vacuously true since $\alpha$ has no initial segment.

**Inductive Step:** Let $\alpha$ and $\beta$ have the property. it suffices to show that $(\alpha \vee \beta)$ also has the property. We proceed by cases.

Case 1: $\gamma = ($ , clearly the property holds.

Case 2: $\gamma = (\gamma'$ where $\gamma'$ is an initial segment of $\alpha$. By our inductive hypothesis, the property holds.

Case 3: $\gamma = (\alpha$. Then $\alpha$ has balanced right and left bracket. Adding one left bracket ensures that the property holds.

Case 4: $\gamma = (\alpha \vee \gamma''$ where $\gamma''$ is an initial segment of $\beta$. By our inductive hypothesis, the property holds.

Case 5: $\gamma = (\alpha \vee \beta$ . Then $\alpha$ and $\beta$ have balanced right and left bracket. Adding one left bracket ensures that the property holds.

Hence, by induction, for any initial segment of $(\alpha \vee \beta)$, then the number of right brackets is strictly less that the number of left brackets. $\square$

# 4 The Semantics of Propositional Logic

The semantics of propositional logic rests on two truth values $\{T, F\}$.

**Truth Assignment:** A truth assignment is a function $V : S \to \{T, F\}$ where $S$ is the set of propositional variables.

We define how to extend such an assignment to assign truth values to every propositional formula over $S$. Here we define $\bar{V}$ : Form $\to \{T, F\}$ by four fixed truth tables, each for $\neg, \vee, \wedge, \to$.

| $\varphi$ | $\psi$ | $\neg\varphi$ | $\varphi \wedge \psi$ | $\varphi \vee \psi$ | $\varphi \to \psi$ |
|---|---|---|---|---|---|
| $T$ | $T$ | $F$ | $T$ | $T$ | $T$ |
| $T$ | $F$ | $F$ | $F$ | $T$ | $F$ |
| $F$ | $T$ | $T$ | $F$ | $T$ | $T$ |
| $F$ | $F$ | $T$ | $F$ | $F$ | $T$ |

Formally we define $\bar{V}$ recursively:

1. If $\varphi$ is an atomic formula, $\bar{V}(\varphi) = V(\varphi)$.
2. If $\varphi = \neg\psi$ for some formula $\psi$ then
$$\bar{V}(\varphi) = \begin{cases} F, & \text{if } V(\psi) = T \\ T, & \text{otherwise} \end{cases}$$
3. If $\varphi = (\psi \wedge \chi)$ for some formulas $\psi, \chi$, then
$$\bar{V}(\varphi) = \begin{cases} T, & \text{if } \bar{V}(\psi) = T \text{ and } \bar{V}(\chi) = T, \\ F, & \text{otherwise.} \end{cases}$$
4. If $\varphi = (\psi \vee \chi)$ for some formulas $\psi, \chi$, then
$$\bar{V}(\varphi) = \begin{cases} T, & \text{if } \bar{V}(\psi) = T \text{ or } \bar{V}(\chi) = T, \\ F, & \text{otherwise.} \end{cases}$$
5. If $\varphi = (\psi \to \chi)$ for some formulas $\psi, \chi$, then
$$\bar{V}(\varphi) = \begin{cases} F, & \text{if } \bar{V}(\psi) = T \text{ and } \bar{V}(\chi) = F, \\ T, & \text{otherwise.} \end{cases}$$

But why is $\bar{V}$ well-defined?

**Lemma 4.1.** *For every wff $\alpha$, one and only one of the following statements hold:*

1. *$\alpha$ is a propositional variable*
2. *$\alpha = (\neg\beta)$ for some wff $\beta$*
3. *$\alpha = (\beta \vee \gamma)$ for wffs $\beta$ and $\gamma$*
4. *$\alpha = (\beta \wedge \gamma)$ for wffs $\beta$ and $\gamma$*
5. *$\alpha = (\beta \to \gamma)$ for wffs $\beta$ and $\gamma$*

*Proof.* We prove both existence and uniqueness by induction on the *length* of the wff $\alpha$.

**Existence:** By definition, wffs are built recursively:

- Any propositional variable is a wff.
- If $\beta$ and $\gamma$ are wffs, then $(\neg\beta), (\beta \vee \gamma), (\beta \wedge \gamma), (\beta \to \gamma)$ are wffs.

Hence every wff falls into one of the five cases above.

**Uniqueness:** Assume $\alpha$ admits two decompositions:

$$\alpha = (\text{some decomposition using one connective}) = (\text{another decomposition using possibly another connective}).$$

We argue by induction on the length $|\alpha|$:

- If $\alpha$ is atomic, there is nothing to prove; the decomposition is trivially unique.
- If $\alpha$ is non-atomic, it begins with a left parenthesis ( and ends with a right parenthesis ). Scan $\alpha$ from left to right:
  - Count left and right parentheses. The *outermost connective* occurs at the unique position where the number of left and right parentheses balances after the first symbol.
  - This identifies the main connective (either $\neg, \vee, \wedge, \to$) and uniquely splits $\alpha$ into immediate subformula(s) $\beta$ and possibly $\gamma$.

  By the induction hypothesis, these subformulas themselves have unique decompositions. Therefore, the decomposition of $\alpha$ is unique.

Hence, for every wff $\alpha$, exactly one of the five forms holds and the subformulas are uniquely determined. □

**Theorem 4.1.** *For every truth assignment $V$ and every wff $\alpha$, there is only one value for $\bar{V}(\alpha)$ that is consistent with our definition of $\bar{V}$.*

*Proof.* We proceed by induction.

**Base Case:** If $\alpha$ is a propositional variable, then $\bar{V}(\alpha) = V(\alpha)$ hence unique.

**Inductive Step:** Let $\alpha = (\beta \vee \gamma)$. By the previous lemma, there is no other way to write $\alpha$. Hence $\bar{V}(\alpha)$ is unique since both $\bar{V}(\beta)$ and $\bar{V}(\gamma)$ are unique (inductive hypothesis). □

Some basic semantic notions:

- $\alpha$ is satisfiable if there exists some truth assignment $V$, such that $\bar{V}(\alpha) = T$.

- $\alpha$ is a tautology if for every truth assignment $V$, $\bar{V}(\alpha) = T$.
- $\alpha$ is a contradiction if for every truth assignment $V$, $\bar{V}(\alpha) = F$.

The above notions involve only a single formula. The following will involve multiple formulas:

- $\alpha$ logically implies $\beta$ (written $\alpha \models \beta$) if, for every truth assignment $V$, if $\bar{V}(\alpha) = T$ then also $\bar{V}(\beta) = T$.
- $\alpha$ and $\beta$ are logically equivalent if they logically imply each other.

Next, we discuss semantic notions on sets of formulas. Let $\Sigma$ denote a set of propositional formulas.

- $\Sigma$ is satisfiable if there exists a truth assignment $V$ such that for every $\alpha \in \Sigma$, $\bar{V}(\alpha) = T$.
- $\Sigma$ logically implies $\alpha$ (written $\Sigma \models \alpha$) if for every truth assignment $V$ and for all $\beta \in \Sigma$, if $\bar{V}(\beta) = T$ then $\bar{V}(\alpha) = T$.

**Claim:** If $\Sigma$ contains a contradiction, then for every formula $\alpha$, $\Sigma$ logically implies $\alpha$.

*Proof.* Vacuously true. $\square$

The key idea here is that if we start by assuming a contradiction, then we can imply whatever we want. Bertrand Russel's famous proof that $2 + 2 = 5$ implies that he is the pope is a good example for this.

Our definition of propositional logic was based on the connectives $\langle \neg, \vee, \wedge, \to \rangle$. But how general does it make our logic? Could we have chosen better connectives?

A **truth functional** is a function $f : \{T, F\}^n \to \{T, F\}$ for some $n$. Each of our four truth tables therefore is a truth functional. But we can think of truth functional that don't directly map to our four connectives. Consider the connective NAND($\uparrow$). We can also consider the function $\text{Maj}_n$ where for every odd number $n$, it returns the majority truth value. So, our logical connectives are a part of larger set of truth functionals.

**Definition 4.1.** *A set of truth functionals is called adequate if propositional formulas over that set can express every possible truth functional.*

**Theorem 4.2.** *The set* $\{\neg, \wedge, \vee, \to\}$ *is adequate.*

*Proof.* Let $f$ be any truth function of $n$ variables $p_1, \ldots, p_n$. Consider its truth table. For each row $i$ in which $f$ takes the value $\mathbf{T}$, define a monomial $\alpha_i$ as follows:

$$\alpha_i = \bigwedge_{j=1}^{n} \beta_{ij},$$

where

$$\beta_{ij} = \begin{cases} p_j & \text{if } p_j \text{ is } \mathbf{T} \text{ in row } i, \\ \neg p_j & \text{if } p_j \text{ is } \mathbf{F} \text{ in row } i. \end{cases}$$

Each $\alpha_i$ is true exactly on row $i$ and false on all other rows. Now define

$$\varphi = \bigvee_{i:\, f(i)=\mathbf{T}} \alpha_i.$$

Then $\varphi$ is true exactly on those rows where $f$ is true, and false otherwise. Hence $\varphi$ expresses $f$.

Since $\varphi$ is built using only $\neg$, $\wedge$, and $\vee$, and $\to$ can be defined from these, it follows that every truth function is expressible using $\{\neg, \wedge, \vee, \to\}$. Therefore, this set is adequate. $\square$

There are different ways of showing whether a set of logical symbols are adequate or not.

**Claim:** The set $\{\wedge, \vee, \to\}$ is not adequate.

*Proof.* Let $V_T$ be the valuation assigning $\mathbf{T}$ to every propositional variable. We show by structural induction that for every formula $\varphi$ built from $\{\wedge, \vee, \to\}$,

$$V_T(\varphi) = \mathbf{T}.$$

**Base Case:** If $\varphi$ is a propositional variable, then $V_T(\varphi) = \mathbf{T}$ by definition of $V_T$.

**Inductive Step:** Suppose $V_T(\alpha) = \mathbf{T}$ and $V_T(\beta) = \mathbf{T}$. Then

$$V_T(\alpha \wedge \beta) = \mathbf{T}, \quad V_T(\alpha \vee \beta) = \mathbf{T}, \quad V_T(\alpha \to \beta) = \mathbf{T}.$$

Thus the property is preserved under all connectives in $\{\wedge, \vee, \to\}$.

Hence every formula $\varphi$ in this language satisfies $V_T(\varphi) = \mathbf{T}$.

Now consider a truth function $f$ such that $f(\mathbf{T}, \ldots, \mathbf{T}) = \mathbf{F}$. No formula built from $\{\wedge, \vee, \to\}$ can express $f$, since all such formulas evaluate to $\mathbf{T}$ under $V_T$.

Therefore, $\{\wedge, \vee, \to\}$ is not adequate. $\square$

**Claim:** The set $\{\neg, \vee\}$ is adequate.

*Proof.* We show that $\wedge$ can be defined using only $\neg$ and $\vee$. By De Morgan's law,

$$\alpha \wedge \beta \equiv \neg(\neg\alpha \vee \neg\beta).$$

Thus every occurrence of $\wedge$ can be eliminated in favor of $\neg$ and $\vee$.

Since $\{\neg, \wedge, \vee\}$ is adequate, it follows that every truth function can be expressed using only $\neg$ and $\vee$. Hence $\{\neg, \vee\}$ is adequate. $\square$

Further semantic notions on sets of propositions $\Sigma$:

- $\Sigma$ is complete if for every propositional formula $\varphi$, $\Sigma \models \varphi$ or $\Sigma \models \neg\varphi$.
- $\Sigma$ is both complete and satisfiable if for every propositional formula $\varphi$, $\Sigma \models \varphi$ or $\Sigma \models \neg\varphi$ but not both.
- $\Sigma$ is maximally satisfiable if $\Sigma$ is satisfiable and for every $\varphi$, if $\Sigma \not\models \varphi$ ($\varphi \notin \Sigma$), then $\Sigma \cup \{\varphi\}$ is not satisfiable.

**Theorem 4.3.** *For every set of propositional formulas $\Sigma$, tfae:*

1. *$\Sigma$ is complete and satisfiable.*
2. *$\Sigma$ is maximally satisfiable.*

*Proof.* ($1 \to 2$) Let $\Sigma$ be complete and satisfiable. Given $\varphi$, if $\Sigma \not\models \varphi$, by completeness, we have $\Sigma \models \neg\varphi$. Therefore, $\Sigma \cup \{\varphi\}$ is not satisfiable. Hence $\Sigma$ is maximally satisfiable.

($2 \to 1$) Let $\Sigma$ be maximally satisfiable. Then if $\Sigma \not\models \varphi$, then $\Sigma \cup \{\varphi\}$ is not satisfiable. Hence $\Sigma \models \neg\varphi$. □

**Claim:** $\Sigma$ is maximally satisfiable if and only if there is exactly one $V$ that satisfies all formulas in $\Sigma$.

*Proof.* ($\to$) Let $\Sigma$ be maximally satisfiable. Then clearly there exists at least one $V$ that satisfies $\Sigma$. We must now show that $V$ is unique. Let $V$ and $V'$ be two assignments that satisfy $\Sigma$. If they are different, then there exists a propositional variable $p$ such that $V(p) = T$ and $V'(p) = F$. Since $V'(p) = F$, $\Sigma \cup \{\neg p\}$ is satisfiable. But since $V(p) = T$, $\Sigma \cup \{\neg p\}$ is unsatisfiable. Hence, a contradiction.

($\leftarrow$) Let $V$ be the unique truth assignment that satisfies all formulas in $\Sigma$. Then clearly, if $\Sigma \not\models \alpha$ then $\Sigma \cup \{\alpha\}$ is not satisfiable since $V$ is the only assignment that satisfies $\Sigma$ and $\alpha$ is not satisfied by $V$. Hence, $\Sigma$ is maximally satisfiable. □

**Exercise 4.0**

(a) Is $(((P \to Q) \to P) \to P)$ a tautology?
(b) Define $\sigma_k$ recursively as follows: $\sigma_0 = (P \to Q)$ and $\sigma_{k+1} = (\sigma_k \to P)$. For which values of $k$ is $\sigma_k$ a tautology? (Part (a) corresponds to $k = 2$.)
**Solution:**
(a) Let $\varphi = (((P \to Q) \to P) \to P)$. Assume $\varphi$ is not a tautology. Then $\varphi = F$ only when $P = F$. Regardless of what value we assign $Q$, we see that $\varphi$ evaluates to true. Hence, a contradiction. Thus, $\varphi$ is a tautology.
(b) Assuming $k \in \mathbb{N}\backslash\{0\}$, then we can see that $\sigma_k$ is a tautology whenever $k$ is even.

**Exercise 4.1**

Show that the following hold:
(a) $\Sigma; \alpha \models \beta$ iff $\Sigma \models (\alpha \to \beta)$.
(b) $\big((\alpha \models \beta) \wedge (\beta \models \alpha)\big)$ iff $\models (\alpha \leftrightarrow \beta)$.
(Recall that $\Sigma; \alpha = \Sigma \cup \{\alpha\}$, the set together with the one possibly new member $\alpha$.)
**Solution:**
(a) ($\Rightarrow$) Let $\Sigma; \alpha \models \beta$. Then for any truth assignment $V$, if $V(\Sigma) = T$ and $\bar{V}(\alpha) = T$, then $\bar{V}(\beta) = T$. Thus, whenever $V(\Sigma) = T$, then $V((\alpha \to \beta)) = T$ as well. Hence, $\Sigma \models (\alpha \to \beta)$.
($\Leftarrow$) Let $\Sigma \models (\alpha \to \beta)$. For $V(\Sigma) = T$, we have $V((\alpha \to \beta)) = T$. Therefore, whenever $V(\alpha) = T$, $V(\beta) = T$ as well. Hence, $\Sigma; \alpha \models \beta$ holds.

(b) ($\Rightarrow$) Let $((\alpha \models \beta) \wedge (\beta \models \alpha))$. Then this basically telling us that $\alpha$ and $\beta$ have the same truth value under any evaluation. Therefore, $(\alpha \leftrightarrow \beta)$ is a tautology. Hence, $\models (\alpha \leftrightarrow \beta)$ holds.
($\Leftarrow$) If $(\alpha \leftrightarrow \beta)$ is a tautology, then $\alpha$ and $\beta$ have the same truth value under any evaluation $V$. Hence $((\alpha \models \beta) \wedge (\beta \models \alpha))$ holds.

**Exercise 4.2**

Prove or refute each of the following assertions:
(a) If either $\Sigma \models \alpha$ or $\Sigma \models \beta$, then $\Sigma \models (\alpha \vee \beta)$.
(b) If $\Sigma \models (\alpha \vee \beta)$, then either $\Sigma \models \alpha$ or $\Sigma \models \beta$.
**Solution:**
(a) If $\Sigma \models \alpha$ or $\Sigma \models \beta$ then for any truth assignment $V$ that is true for $\Sigma$, either $V(\alpha) = T$ or $V(\beta) = T$. Hence $V((\alpha \vee \beta)) = T$. Thus, $\Sigma \models (\alpha \vee \beta)$.
(b) The converse doesn't hold. As a counter example, we know that $\models (\alpha \vee (\neg\alpha))$. But, neither $\models \alpha$ nor $\models (\neg\alpha)$ need not hold.

**Exercise 4.3**

(a) Show that if $v_1$ and $v_2$ are truth assignments which agree on all the sentence symbols in the wff $\alpha$, then $\overline{v}_1(\alpha) = \overline{v}_2(\alpha)$. Use the induction principle.

(b) Let $S$ be a set of sentence symbols that includes those in $\Sigma$ and $\tau$ (and possibly more). Show that $\Sigma \models \tau$ iff every truth assignment for $S$ which satisfies every member of $\Sigma$ also satisfies $\tau$. (This is an easy consequence of part (a). The point of part (b) is that we do not need to worry about getting the domain of a truth assignment exactly perfect, as long as it is big enough. For example, one option would be always to use truth assignments on the set of all sentence symbols. The drawback is that these are infinite objects, and there are a great many — uncountably many — of them.)

**Solution:**

(a) We proceed by structural induction on $\alpha$.

**Base Case:** Let $\alpha = p$ be a propositional variable. Since $v_1$ and $v_2$ agree on all sentence symbols in $\alpha$, we have

$$\bar{v}_1(p) = v_1(p) = v_2(p) = \bar{v}_2(p).$$

**Inductive Step:**

*Case 1:* $\alpha = \neg\psi$. All sentence symbols of $\psi$ occur in $\alpha$, so $v_1$ and $v_2$ agree on them. By the inductive hypothesis,

$$\bar{v}_1(\psi) = \bar{v}_2(\psi).$$

Hence,

$$\bar{v}_1(\neg\psi) = \neg\bar{v}_1(\psi) = \neg\bar{v}_2(\psi) = \bar{v}_2(\neg\psi).$$

*Case 2:* $\alpha = (\psi \vee \phi)$. All sentence symbols of $\psi$ and $\phi$ occur in $\alpha$, so $v_1$ and $v_2$ agree on them. By the inductive hypothesis,

$$\bar{v}_1(\psi) = \bar{v}_2(\psi), \quad \bar{v}_1(\phi) = \bar{v}_2(\phi).$$

Thus,

$$\bar{v}_1(\psi \vee \phi) = \bar{v}_1(\psi) \vee \bar{v}_1(\phi) = \bar{v}_2(\psi) \vee \bar{v}_2(\phi) = \bar{v}_2(\psi \vee \phi).$$

Therefore, $\bar{v}_1(\alpha) = \bar{v}_2(\alpha)$ for all $\alpha$.

(b) ($\Rightarrow$) Suppose $\Sigma \models \tau$. Let $v$ be any truth assignment on $S$ such that $v(\Sigma) = T$. Let $v'$ be the restriction of $v$ to the sentence symbols occurring in $\Sigma \cup \{\tau\}$. Then $v'$ is a truth assignment (on a smaller set) with $v'(\Sigma) = T$. Since $\Sigma \models \tau$, we have $\bar{v}'(\tau) = T$.
By part (a), $v$ and $v'$ agree on all sentence symbols in $\tau$, hence

$$\bar{v}(\tau) = \bar{v}'(\tau) = T.$$

Thus, every truth assignment on $S$ that satisfies $\Sigma$ also satisfies $\tau$.
($\Leftarrow$) Suppose every truth assignment on $S$ that satisfies $\Sigma$ also satisfies $\tau$. Let $v$ be any truth assignment (on the sentence symbols of $\Sigma \cup \{\tau\}$) such that $v(\Sigma) = T$. Extend $v$ to a truth assignment $w$ on $S$ arbitrarily. Then $w(\Sigma) = T$, so by assumption $w(\tau) = T$.
By part (a), since $v$ and $w$ agree on all sentence symbols in $\tau$, we have

$$\bar{v}(\tau) = \bar{w}(\tau) = T.$$

Hence, $\Sigma \models \tau$.

## 5 Compactness Theorem

**Fact:** $\Sigma \models \alpha$ if and only if $\Sigma \cup \{\neg\alpha\}$ is not satisfiable.

**Theorem 5.1** (Finite Models)**.** *For any $\Sigma \models \alpha$, there exists $\Sigma_0 \subseteq \Sigma$ where $\Sigma_0$ is finite, such that $\Sigma_0 \models \alpha$.*

**Intuition:** Look at the truth table for $\alpha$. If $\alpha$ has $n$ propositional variables, then the truth table will have $2^n$ rows. We'll be focusing on those rows where $\alpha$ is false. Let $V_i$ be the $i$-th truth assignment for the false rows. Consider $\Sigma$ under $V_i$. Because $\Sigma \models \alpha$, we know that there exists some $\beta_i \in \Sigma$ such that $V_i(\beta_i) = F$. Define $\Sigma_0 = \{\beta_i : \text{ for all } i\}$. We now claim that $\Sigma_0 \models \alpha$. It suffices to show that $\Sigma_0 \cup \{\neg\alpha\}$ is not satisfiable. Assume, for contradiction, that it is. Hence, there exists a truth assignment $V$ that satisfies it, i.e $V(\neg\alpha) = T$ and $V(\alpha) = F$. Hence, there exists an $i$ such that $V_i = V$. But since $V$ satisfies $\Sigma_0$, $V(\beta_i) = T$ for all $\beta_i \in \Sigma_0$. This is clearly a contradiction. Hence, $\Sigma_0 \cup \{\neg\alpha\}$ is not satisfiable. Therefore, $\Sigma_0 \models \alpha$.

*Proof.* Let $\alpha$ have $n$ propositional variables, and consider all $2^n$ truth assignments. Focus on those assignments $V_1, \ldots, V_k$ under which $\alpha$ is false.

Since $\Sigma \models \alpha$, each $V_i$ fails to satisfy $\Sigma$. Hence, for each $i$ there exists $\beta_i \in \Sigma$ such that $V_i(\beta_i) = F$. Let

$$\Sigma_0 := \{\beta_1, \ldots, \beta_k\}.$$

Then $\Sigma_0$ is finite.

We claim that $\Sigma_0 \models \alpha$. It suffices to show that $\Sigma_0 \cup \{\neg\alpha\}$ is unsatisfiable. Suppose, for contradiction, that there exists a valuation $V$ such that

$$V \models \Sigma_0 \cup \{\neg\alpha\}.$$

Then $V(\alpha) = F$, so $V = V_i$ for some $i$. But $V \models \Sigma_0$ implies $V(\beta_i) = T$, contradicting $V_i(\beta_i) = F$.

Therefore, $\Sigma_0 \cup \{\neg\alpha\}$ is unsatisfiable, and hence $\Sigma_0 \models \alpha$. □

**Definition 5.1.** *$\Sigma$ is finitely satisfiable (fs) if and only if for all finite $\Sigma_0$ such that $\Sigma_0 \subseteq \Sigma$, $\Sigma_0$ is also satisfiable.*

**Theorem (The Compactness Theorem):** $\Sigma$ is satisfiable if and only if $\Sigma$ is finitely satisfiable.

**Proof Outline:** $(\rightarrow)$ is trivial. If $\Sigma$ is finite, then $\Sigma \subseteq \Sigma$, hence $(\leftarrow)$ also becomes trivial. Its only important to consider the case where $\Sigma$ is infinite. We first extend $\Sigma$ to $\Delta$ such that $\Delta \supseteq \Sigma$ , $\Delta$ is finitely satisfiable, and for all formulas $\alpha$, either $\alpha \in \Delta$ or $\neg\alpha \in \Delta$. We then find a $V$ that satisfies $\Delta$. Hence $V$ satisfies $\Sigma$.

**Lemma 5.1.** *There exists some ordering of all wffs such that every wff has a unique index.*

*Proof.* Let $\Sigma_0 := \{\wedge, \vee, \neg, \rightarrow, (, )\} \cup \mathbf{A}$ be the set of symbols, where $\mathbf{A}$ is the set of propositional variables. Since $\Sigma_0$ is countable, fix an injective function

$$f : \Sigma_0 \rightarrow \mathbb{N}.$$

For any wff $\alpha = x_1 x_2 \cdots x_n$ (a finite string of symbols from $\Sigma_0$), define its code by

$$\ulcorner\alpha\urcorner := p_1^{f(x_1)} p_2^{f(x_2)} \cdots p_n^{f(x_n)},$$

where $p_i$ is the $i$-th prime number.

By the Fundamental Theorem of Arithmetic, every natural number has a unique prime factorization. Hence distinct strings (and therefore distinct wffs) receive distinct codes. Thus this defines an injective mapping from the set of all wffs into $\mathbb{N}$.

Finally, ordering wffs by the natural order of their codes yields the desired enumeration. $\square$

**Lemma 5.2.** *If $\Sigma$ is finitely satisfiable, then $\Sigma \cup \{\alpha\}$ or $\Sigma \cup \{\neg\alpha\}$ are finitely satisfiable.*

*Proof.* Assume both $\Sigma \cup \{\alpha\}$ and $\Sigma \cup \{\neg\alpha\}$ are not finitely satisfiable.Then there exist finite subsets $\Sigma_0 \subseteq \Sigma \cup \{\alpha\}$ and $\Sigma_1 \subseteq \Sigma \cup \{\neg\alpha\}$ such that $\Sigma_0$ and $\Sigma_1$ are unsatisfiable. Define $\Sigma' = \Sigma_0 \cup \Sigma_1 \backslash \{\alpha, \neg\alpha\}$. Then, $\Sigma'$ is finite and $\Sigma' \subseteq \Sigma$.Hence, $\Sigma'$ must be satisfiable. This means there exists some $V$ such that $V$ satisfies $\Sigma'$. For $\alpha$, either $V(\alpha) = T$ or $V(\neg\alpha) = T$. If $V(\alpha) = T$, then $\Sigma_0$ is satisfiable. If $V(\neg\alpha) = T$, then $\Sigma_1$ is satisfiable. In both cases, we get a contradiction. Hence, either $\Sigma \cup \{\alpha\}$ or $\Sigma \cup \{\neg\alpha\}$ must be fs. $\square$

We'll now restate the compactness theorem and prove it.

**Theorem 5.2** (Compactness Theorem)**.** *$\Sigma$ is satisfiable if and only if $\Sigma$ is finitely satisfiable.*

*Proof.* ($\Rightarrow$) Suppose $\Sigma$ is satisfiable. Then there exists a valuation $V$ such that $V \models \Sigma$. Hence $V \models \Sigma_0$ for every finite $\Sigma_0 \subseteq \Sigma$, so every finite subset is satisfiable.

($\Leftarrow$) We construct $\Delta$ inductively. Start with $\Delta_0 = \Sigma$. By our first lemma, we can index our wff as $\alpha_i$.For all $i$, we define $\Delta_{i+1}$ as:

$$\Delta_{i+1} = \begin{cases} \Delta_i \cup \{\alpha_i\} \text{ if it is finitely satisfiable,} \\ \Delta_i \cup \{\neg\alpha_i\} \text{ otherwise.} \end{cases}$$

Our second lemma ensures that all $\Delta_i$ is finitely satisfiable. Finally, we define $\Delta$ as:

$$\Delta = \bigcup_{i \geqslant 0} \Delta_i$$

The following facts are observed:

- $\Sigma \subseteq \Delta$
- For any wff $\alpha$, either $\alpha \in \Delta$ or $\neg\alpha \in \Delta$
- $\Delta$ is finitely satisfiable, since any finite subset of $\Delta$ is a finite subset of some $\Delta_i$. And since $\Delta_i$ is finitely satisfiable, any finite subset of $\Delta$ is thus satisfiable.

Finally we must find a $V$ that satisfies $\Delta$. For all propositional variables $p \in \mathbf{A}$, define $V$ as;

$$V(p) = \begin{cases} T, \text{if } p \in \Delta \\ F, \text{if } p \notin \Delta \end{cases}$$

We claim that $V(\alpha) = T \iff \alpha \in \Delta$. We prove this by structural induction.

**Base Case:** For any propositional variable $p$, $V(p) = T \iff p \in \Delta$, by definition.

**Inductive Step:** Assume $\alpha$ and $\beta$ satisfy our property. Let $\gamma = (\neg\alpha)$. Then $V(\gamma) = T \iff V(\alpha) = F \iff \alpha \notin \Delta \iff (\neg\alpha) \in \Delta \iff \gamma \in \Delta$.

Let $\gamma = (\alpha \wedge \beta)$. Then $V(\gamma) = T \iff V(\alpha) = T$ and $V(\beta) = T \iff \alpha, \beta \in \Delta \iff (\alpha \wedge \beta) \in \Delta \iff \gamma \in \Delta$.

Thus, $V$ satisfies $\Delta$ and hence, satisfies $\Sigma$.

Therefore, $\Sigma$ is satisfiable. □

# 6 Consequences of the Compactness Theorem

Consider two infinite sets: the set of all propositional variables and the set of all truth assignments. Intuition tells us that the later set is bigger than the former (we'll explore this later). But for now, we want to focus on the relationship between these two sets.

Consider $\Sigma$ to be a subset of the set of all propositional variables. For any such $\Sigma$, we define $A_\Sigma$ which is a subset of the set of all truth assignments such that:

$$A_\Sigma = \{V : V \text{ satisfies } \Sigma\}$$

We can use this to demonstrate the expressive power of propositional logic. For a moment, let us consider an undirected graph $G = (V, E)$. A graph coloring by $k$ colors is a function $c : V \to \{1, \ldots, k\}$ such that for all $(x, y) \in E$, $c(x) \neq c(y)$. A graph is $k$-colorable if there exists a coloring of the graph with $k$ colors. Given a graph $G = (V, E)$, we can express its colorability with propositional logic as follows:

Let $V = \{v_1, \ldots, v_n\}$, we will use formulas over $p_1, \ldots, p_n$. We must find a set of formulas $\Sigma$ over $p_1, \ldots, p_n$ such that $\Sigma$ is satisfiable if and only if $G$ is 2-colorable. We define it as follows:

$$\Sigma_G = \{((p_i \to \neg p_j) \wedge (\neg p_i \to p_j)) : (v_i, v_j) \in E\}$$

**Claim:** $G$ is two colorable if and only if $\Sigma_G$ is satisfiable.

*Proof.* ($\Rightarrow$) Let $G$ be two colorable. Then we define a truth assignment $\mathfrak{T}_c$ such that:

$$\mathfrak{T}_c(p_i) = \begin{cases} T, & \text{if } c(v_i) = 1 \\ F, & \text{if } c(v_i) = 2 \end{cases}$$

We now check if $\mathfrak{T}_c$ satisfies $\Sigma_G$. Note in $\Sigma_G$, $V(p_i) \neq V(p_j)$ whenever $(v_i, v_j) \in E$, i.e., no adjacent variables have the same truth value. Hence, we can see that $\mathfrak{T}_c$ satisfies $\Sigma_G$.

($\Leftarrow$) Let $\mathfrak{T}$ satisfy $\Sigma_G$. We define a coloring $c$ as follows:

$$c(v_i) = \begin{cases} 1, & \text{if } \mathfrak{T}(p_i) = T \\ 2, & \text{if } \mathfrak{T}(p_i) = F \end{cases}$$

□

With this (and the compactness theorem), we can infer non-trivial results in graph theory. For example:

**Corollary:** For every graph $G = (V, E)$, $G$ is 2-colorable if and only if every finite subgraph of $G$ is 2-colorable.

Since we have successfully "mapped" satisfiability to 2-colorability, we can apply the theorems we know in propositional logic to obtain non-trivial results.

Lets return to $\Sigma$ and $A_\Sigma$. Given a $\Sigma$ we can easily find a corresponding $A_\Sigma$. But is the reverse possible? Recall earlier that we said the set of all truth assignments is larger than the set of all formulas.. And since the relation between $\Sigma$ and $A_\Sigma$ is one to one, there are some subsets of truth assignments that don't correspond to any subset of formulas. But we can prove this fact without invoking set theory.

**Lemma 6.1.** *For every $i$, let $V_i$ be the truth assignment defined by:*

$$V_i(p_j) = \begin{cases} T, & \text{if } j \leqslant i \\ F, & \text{if } j > i \end{cases}$$

*There exists no set of propositional formulas $\Sigma$ such that $A_\Sigma = \{V_i : i \in \mathbb{N}\}$.*

*Proof.* Assume $\Sigma$ is a set of formulas satisfied by each $V_i$ and no other $V$. Let $\Sigma' = \Sigma \cup \{p_i\}_{i \in \mathbb{N}}$. If $\Sigma'$ is satisfiable, then $V_T$ (all true assignment) satisfies $\Sigma$. However, by our assumption on $\Sigma$, $V_T$ shouldn't satisfy it. Hence, $\Sigma'$ is not satisfiable.

Recall that every finite subset of $\Sigma'$ contains finite set of $p_i$s and/or a finite subset of $\Sigma$, hence can be satisfied with $V_i$. Thus, $\Sigma'$ is finitely satisfiable, hence satisfiable. Thus, we have a contradiction. $\square$

**Definition 6.1.** *A set $W$ of propositional truth assignments is definable if there exists a set $\Sigma$ of formulas such that:*

$$W = \{V : V \text{ satisfies } \Sigma\}$$

When $\Sigma$ grows, $W$ shrinks.

Lemma 6.1 shows us that not all $W$s are definable.

# 7 Proof Syntax in PL

We now make a brief return to syntax. We want to build a proof system for propositional logic.

Desirable properties of proof systems:

1. It can only prove "correct" statements.
2. There is a "method" to check whether a given object is a proof.
3. Every correct statement has a proof.

The existence of such a proof system depends on our notion of "correct" and the language we're working in.

In PL, "correct" stands for a tautology. Hence, a proof of $\alpha$ will be a truth table that has all $T$s in the right most column. But this proof system hardly applies to any branch of mathematics. So we need a better system.

Proofs in mathematics typically have two components: Axioms and deduction rules. Our system will be mirroring this. We'll be relying on the symbols $\{\neg, \rightarrow\}$ for simplicity.

**Axioms:** Any wff of one of the following forms:

P1 $(\alpha \rightarrow (\beta \rightarrow \alpha))$

P2 $\Big((\alpha \rightarrow (\beta \rightarrow \gamma)) \rightarrow ((\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \gamma))\Big)$

P3 $\Big(((\neg\alpha) \rightarrow (\neg\beta)) \rightarrow (\beta \rightarrow \alpha)\Big)$

Each of these axioms is a tautology. But not the reverse, i.e, there are tautologies which aren't axioms ($(p \rightarrow p)$, for example).

**Deduction Rules:**

1. Modus Ponens:

$$\frac{\alpha \quad (\alpha \rightarrow \beta)}{\beta}$$

This is the only deduction rule we need. Therefore, the set of provable wffs is the inductive set defined as $I$(Axioms, Modus Ponens). A given proposition is a theorem if it belongs to this set.

A formal proof is a construction sequence in the inductive set. That is, $\alpha_1 \ldots \alpha_n$ is a formal proof of some formula $\beta$ if each $\alpha_i$ is either an axiom or the result of applying the Modus Ponens to some $\alpha_j, \alpha_k$ where $j, k < i$.

**Example:** Show that the statement $(\alpha \rightarrow \alpha)$ is provable.

1. $(\alpha \rightarrow ((\alpha \rightarrow \alpha) \rightarrow \alpha))$ - P1
2. $(\alpha \rightarrow ((\alpha \rightarrow \alpha) \rightarrow \alpha)) \rightarrow ((\alpha \rightarrow (\alpha \rightarrow \alpha)) \rightarrow (\alpha \rightarrow \alpha)))$ - P2
3. $((\alpha \rightarrow (\alpha \rightarrow \alpha)) \rightarrow (\alpha \rightarrow \alpha))$ - Modus Ponens (1,2)
4. $(\alpha \rightarrow (\alpha \rightarrow \alpha))$ - P1
5. $(\alpha \rightarrow \alpha)$ - Modus Ponens (3,4)

**Notation:** We use $\vdash$ to denote provability. $\vdash \alpha$ corresponds to "$\alpha$ has a formal proof".

**Theorem 7.1** (Soundness Theorem)**.** *For every formula $\alpha$, if $\vdash \alpha$, then $\alpha$ is a tautology.*

*Proof.* We proceed by generalized induction on $I$.

**Base Case:** We check that every axiom is a tautology, which holds by definition.

**Inductive Step:** Assume both $\alpha$ and $(\alpha \rightarrow \beta)$ are tautologies, we must show that $\beta$ is a tautology. For contradiction, assume $\beta$ is not a tautology. Then there is an assignment $\bar{V}$ such that $\bar{V}(\beta) = F$ while $\bar{V}(\alpha) = T$ and $\bar{V}(\alpha \rightarrow \beta) = T$. But if $\bar{V}(\beta) = F$, then $\bar{V}(\alpha \rightarrow \beta) = F$, hence a contradiction. Therefore, $\beta$ must be a tautology. □

**Proofs from Assumptions:** Let $\Sigma$ be a set of formulas. The set of formal theorems under the assumption $\Sigma$ is defined as:

$$\mathsf{Thm}(\Sigma) = I(\text{Axioms} \cup \Sigma, \text{Modus Ponens})$$

$\alpha_1 \ldots \alpha_n$ is a formal proof under $\Sigma$ if for each $\alpha_i$, $\alpha_i$ is an axiom, a member of $\Sigma$, or the result of applying the Modus Ponens to some $\alpha_j, \alpha_k$ where $j, k < i$.

**Example**; For every $\alpha, \beta$ show that $\{\alpha, (\neg\alpha)\} \vdash \beta$.

1. $\neg\alpha \to (\neg\beta \to \neg\alpha)$ - $\mathsf{P1}$
2. $\neg\alpha$ - Assumption
3. $\neg\beta \to \neg\alpha$ - Modus Ponens (1,2)
4. $(\neg\beta \to \neg\alpha) \to (\alpha \to \beta)$ - $\mathsf{P3}$
5. $\alpha \to \beta$ - Modus Ponens (3,4)
6. $\alpha$ - Assumption
7. $\beta$ - Modus Ponens (5,6)

This shows us that if our assumptions involve a contradiction, we can literally prove anything.

**Properties of Proofs from Assumptions**:

1. **Extended Soundness Theorem:** If $\Sigma \vdash \alpha$, then $\Sigma \models \alpha$.
2. **Monotonicity:** For every $\Sigma, \Sigma', \alpha$, if $\Sigma \vdash \alpha$ and $\Sigma \subseteq \Sigma'$, then $\Sigma' \vdash \alpha$. In other words, $\mathsf{Th}(\Sigma) \subseteq \mathsf{Th}(\Sigma')$.

   *Proof.* Since $\Sigma \vdash \alpha$, there is a formal proof of $\alpha$ such that $\alpha_1 \ldots \alpha_n$ where $\alpha_n = \alpha$. Each $\alpha_i$ is an axiom or in $\Sigma$, in which case $\alpha_i \in \Sigma'$, or $\alpha_i$ is the outcome of modus ponens. Therefore, $\alpha_1 \ldots \alpha_n$ is a proof on $\Sigma'$. $\square$

3. **Strong Monotonicity:** If for every $\alpha \in \Sigma'$, $\Sigma \vdash \alpha$, then $\mathsf{Th}(\Sigma') \subseteq \mathsf{Th}(\Sigma)$.
4. **The Deduction Theorem:** For every $\Sigma, \alpha, \beta$, $\Sigma \cup \{\alpha\} \vdash \beta$ iff $\Sigma \vdash (\alpha \to \beta)$.

   *Proof.* ($\Rightarrow$) Let $\Sigma \cup \{\alpha\} \vdash \beta$. We rephrase it as for all $\beta \in \mathsf{Th}(\Sigma \cup \{\alpha\})$, $\Sigma \vdash (\alpha \to \beta)$. We prove by generalizes induction on $\mathsf{Th}(\Sigma \cup \{\alpha\})$.

   **Base Case:** We have three cases.

   *Case 1:* $\beta$ is an axiom. Then the proof sequence

   - $\beta$ - Axiom
   - $\beta \to (\alpha \to \beta)$ - $\mathsf{P1}$
   - $(\alpha \to \beta)$ - Modus Ponens.

   *Case 2:* $\beta$ is in $\Sigma$

   - $\beta$ - Assumption
   - $\beta \to (\alpha \to \beta)$ - $\mathsf{P1}$
   - $(\alpha \to \beta)$ - Modus Ponens.

   *Case 3:* $\beta = \alpha$

   - $\alpha$ - Assumption

- $\alpha \to (\alpha \to \alpha)$ - P1
- $\alpha \to \alpha$ - Modus Ponens

**Inductive Step:** Assume $\Sigma \vdash (\alpha \to \gamma)$ and $\Sigma \vdash (\alpha \to (\gamma \to \delta))$ and show that $\Sigma \vdash (\alpha \to \delta)$. By strong monotonicity, it suffices to show that $\Sigma \cup \{(\alpha \to \gamma), (\alpha \to (\gamma \to \delta)\} \vdash \alpha \to \delta$. The proof sequence is as follows:

- $(\alpha \to (\gamma \to \delta)) \to ((\alpha \to \gamma) \to (\alpha \to \delta))$ -P3
- $(\alpha \to (\gamma \to \delta)$ - Assumption
- $((\alpha \to \gamma) \to (\alpha \to \delta))$ - Modus Ponens
- $(\alpha \to \gamma)$ - Assumption
- $(\alpha \to \delta)$ - Modus Ponens

($\Leftarrow$) Let $\Sigma \vdash (\alpha \to \beta)$. By monotonicity, we know that $\Sigma \cup \{\alpha\} \vdash (\alpha \to \beta)$. Then by strong monotonicity, it suffices to show that $\Sigma \cup \{\alpha, (\alpha \to \beta)\} \vdash \beta$, which easily follows by modus ponens.Hence $\Sigma \cup \{\alpha\} \vdash \beta$.

□

**Application of the deduction theorem:** Show that $\vdash ((\neg(\neg\alpha)) \to \alpha)$

It suffices to show that $\{\neg\neg\alpha\} \vdash \alpha$.

1. $\neg\neg\alpha \to (\neg\neg\neg\neg\alpha \to \neg\neg\alpha)$ - P1
2. $\neg\neg\alpha$ - Assumption
3. $\neg\neg\neg\neg\alpha \to \neg\neg\alpha$ - Modus Ponens (1,2)
4. $(\neg\neg\neg\neg\alpha \to \neg\neg\alpha) \to (\neg\alpha \to \neg\neg\neg\alpha)$ - P3
5. $\neg\alpha \to \neg\neg\neg\alpha$ - Modus Ponens (3,4)
6. $(\neg\alpha \to \neg\neg\neg\alpha) \to (\neg\neg\alpha \to \alpha)$ - P3
7. $(\neg\neg\alpha \to \alpha)$ - Modus Ponens (5,6)
8. $\alpha$ - Modus Ponens (2,7)

**Definition 7.1** (Consistency)**.** *$\Sigma$ is consistent if no $\alpha$ exists such that $\Sigma \vdash \alpha$ and $\Sigma \vdash \neg\alpha$.*

**Definition 7.2.** *A more "syntactic" definition goes something like: $\Sigma$ is consistent if there exists some $\alpha$ such that $\Sigma \nvdash \alpha$. (Recall that we showed that an inconsistent system can prove anything.)*

**Claim:** The two definitions are equivalent.

*Proof.* ($\Rightarrow$) $\Sigma$ is consistent if no $\alpha$ exists such that $\Sigma \vdash \alpha$ and $\Sigma \vdash \neg\alpha$. If $\Sigma \vdash \alpha$, then $\Sigma \nvdash \neg\alpha$ and vice versa. Hence there exists some $\alpha$ such that $\Sigma \nvdash \alpha$.

($\Leftarrow$) Let there exist some $\beta$ such that $\Sigma \nvdash \beta$. If $\Sigma \vdash \alpha$ and $\Sigma \vdash \neg\alpha$ for some $\alpha$, then $\Sigma \vdash \Sigma \cup \{\alpha, \neg\alpha\} \vdash \{\alpha, \neg\alpha\}$. But, $\{\alpha, \neg\alpha\} \vdash \beta$ for all $\beta$. Hence, $\Sigma \vdash \beta$ for all $\beta$; a contradiction. □

Let $\Sigma \subseteq \Sigma'$. If $\Sigma'$ is consistent, then so is $\Sigma$. Hence,$\varnothing$ is consistent (?)

The Soundness theorem states that if $\vdash \alpha$, then $\alpha$ is a tautology. For any propositional variable $p$, $\nvdash p$ since $p$ can not be a tautology. Hence $\varnothing$ is consistent.

**Theorem 7.2.** *Every satisfiable $\Sigma$ is consistent.*

*Proof.* Let $\Sigma$ be satisfiable. If $\Sigma \vdash \alpha$ and $\Sigma \vdash \neg\alpha$. By the soundness theorem, it follows that $\Sigma \models \alpha$ and $\Sigma \models \neg\alpha$. Hence for some truth assignment $\bar{V}$, where $\bar{V}(\Sigma) = T$, $\bar{V}(\alpha) = T$ and $\bar{V}(\neg\alpha) = T$; a contradiction. $\square$

**Definition 7.3.** *We say that $\Sigma$ is maximally consistent if $\Sigma$ is consistent such that for every $\alpha$, either $\Sigma \vdash \alpha$ or $\Sigma \cup \{\alpha\}$ is inconsistent.*

Note that whenever $\Sigma \vdash \alpha$, then $\Sigma \cup \{\alpha\}$ is no stronger that $\Sigma$, i.e., for any $\beta$, if $\Sigma \cup \{\alpha\} \vdash \beta$, then $\Sigma \vdash \beta$.

**Example:** Let $\Sigma = \{p_1\}$ over the variables $\{p_i\}_{i\in\mathbb{N}}$. We claim $\Sigma$ is consistent but not maximally consistent.

It's easy to see that $\Sigma$ is consistent since it is satisfiable. Now, let $\alpha = p_2$. Then $\{p_1\} \not\models p_2$ since we can find a truth assignment that doesn't agree on the two variables. Hence, it follows that $\{p_1\} \nvdash p_2$. Consider now $\{p_1, p_2\}$. This set is still consistent because it is satisfiable. Hence, $\Sigma$ is not maximally consistent.

**Lemma 7.1.** *For every consistent $\Sigma$, there exists a maximally consistent $\Sigma'$ such that $\Sigma \subseteq \Sigma'$.*

*Proof.* Let $\{\alpha_i\}_{i\in\mathbb{N}}$ be a list of all wffs over $\{p_i\}_{i\in\mathbb{N}}$. Let $\Sigma_0 = \Sigma$. We construct a sequence of sets of wffs $\Sigma_0 \subseteq \Sigma_1 \subseteq \ldots \Sigma_n \subseteq \Sigma_{n+1}$ for all $n \in \mathbb{N}$. Such that:

1. Each $\Sigma_i$ is consistent.
2. For every $i$, either $\Sigma_i \vdash \alpha_i$ or $\Sigma_i \vdash \neg\alpha_i$.

We proceed by induction on $i$.

**Base Case:** $\Sigma_0 = \Sigma$. By definition, $\Sigma$ is consistent.

**Inductive Step:** Given $\Sigma_i$, if $\Sigma_i \vdash \neg\alpha_{i+1}$, let $\Sigma_i = \Sigma_{i+1}$. If $\Sigma_i \nvdash \alpha_{i+1}$, let $\Sigma_{i+1} = \Sigma_i \cup \{\alpha_{i+1}\}$. In either case, both 1 and 2 are satisfied.

Now, we have $\Sigma' = \bigcup_{i\in\mathbb{N}} \Sigma_i$. We now claim that $\Sigma'$ is maximally consistent. For every $\alpha$, $\alpha = \alpha_i$ for some $i \in \mathbb{N}$. Hence, by requirements, $\Sigma_i \vdash \alpha_i$ or $\Sigma_i \vdash \neg\alpha_i$. We have $\Sigma_i \subseteq \Sigma'$ for all $i$. By monotonicity, it follows that $\Sigma_i \vdash \alpha_i$ or $\Sigma_i \vdash \neg\alpha_i$.

Assume, for contradiction, $\Sigma'$ is not consistent. Then for some $\alpha$, $\Sigma' \vdash \alpha$ and $\Sigma' \vdash \neg\alpha$. Let $\beta_1, \ldots, \beta_k$ be a formal proof of $\alpha$ from $\Sigma'$ and let $\gamma_1, \ldots, \gamma_m$ be the formal proof of $\neg\alpha$ from $\Sigma'$. Each $\beta_i$ that is an assumption belongs to some $\Sigma_{m_i} \subseteq \Sigma'$. Similarly, each $\gamma_i$ that is an assumption belongs to some $\Sigma_{m_j} \subseteq \Sigma'$. Since both proofs are finite, there is some $i^*$ that is bigger than all $m_i$ and $m_j$. Therefore, each assumptions $\beta_i, \gamma_i \in \Sigma_{i*}$. Hence, $\Sigma_{i*}$ proves both $\alpha$ and $\neg\alpha$, making it inconsistent. This contradicts our original requirement that each $\Sigma_i$ must be consistent. $\square$

**Theorem 7.3.** *Every consistent $\Sigma$ is satisfiable.*

*Proof.* Let $\Sigma'$ be a maximally consistent set of wffs such that $\Sigma \subseteq \Sigma'$. Define a truth assignment $\bar{V}_{\Sigma'}$ as follows:

$$\bar{V}_{\Sigma'}(p_i) = \begin{cases} T \text{ if } \Sigma' \vdash p_i \\ F, \text{ otherwise} \end{cases}$$

We claim for every formula $\alpha$, $\bar{V}_{\Sigma'}(\alpha) = T$ iff $\Sigma' \vdash \alpha$. We prove this by generalized induction on the set of all formulas.

**Base Case:** Let $\alpha = p$. Then, by definition, the desired property holds.

**Inductive Step:** Assume the claim holds for $\beta$ and $\gamma$. We now show it holds for $\neg\beta$ and $\beta \to \gamma$.

*Case 1:* ($\Rightarrow$) For $\neg\beta$, if $\Sigma' \vdash \neg\beta$, by consistency, $\Sigma' \nvdash \beta$. Hence, by the induction hypothesis $\bar{V}_{\Sigma'}(\beta) = F$. Thus $\bar{V}_{\Sigma'}(\neg\beta) = T$. ($\Leftarrow$) Assume $\Sigma' \nvdash \neg\beta$. Then by maximality, $\Sigma' \vdash \beta$. By the induction hypothesis $\bar{V}_{\Sigma'}(\beta) = T$, hence $\bar{V}_{\Sigma'}(\neg\beta) = F$.

*Case 2:* ($\Rightarrow$) For $\beta \to \gamma$, if $\Sigma' \vdash (\beta \to \gamma)$. Then either $\Sigma' \vdash \beta$, which means $\Sigma' \vdash \gamma$. Using the induction hypothesis $\bar{V}_{\Sigma'}(\beta \to \gamma) = T$. Or, $\Sigma' \nvdash \beta$, in which case we have $\bar{V}_{\Sigma'}(\beta \to \gamma) = T$. ($\Leftarrow$) If $\Sigma' \nvdash (\beta \to \gamma)$, then $\Sigma' \nvdash \gamma$, because if it does, we can use the axiom $\gamma \to (\beta \to \gamma)$ and modus ponens to obtain $\Sigma' \vdash (\beta \to \gamma)$. By the induction hypotheses, we have $\bar{V}_{\Sigma'}(\gamma) = F$. For contradiction, assume $\bar{V}_{\Sigma'}(\beta) = F$. Then $\Sigma' \nvdash \beta$, hence, $\Sigma' \vdash \neg\beta$. Hence, $\Sigma' \vdash (\beta \to \gamma)$; a contradiction. Therefore, $\bar{V}_{\Sigma'}(\beta) = T$. ☐

**Theorem 7.4** (Completeness Theorem). *For all $\alpha$ and any set of wffs $\Sigma$, if $\Sigma \models \alpha$, $\Sigma \vdash \alpha$.*

*Proof.* Let $\Sigma \models \alpha$. For contradiction, assume $\Sigma \nvdash \alpha$. It follows then that $\Sigma \cup \{\neg\alpha\}$ is consistent. By our earlier theorem, we see that $\Sigma \cup \{\neg\alpha\}$ is satisfiable. Hence, $\Sigma \not\models \alpha$; a contradiction. ☐

**Syntax and Semantics Relation Summarized**

- $\vdash \alpha \iff \alpha$ is a tautology
- $\Sigma \vdash \alpha \iff \Sigma \models \alpha$
- $\Sigma$ is consistent $\iff$ $\Sigma$ is satisfiable
- $\Sigma$ is maximally consistent $\iff$ $\Sigma$ is maximally satisfiable.

Recall the compactness theorem that states $\Sigma$ is satisfiable if and only if every finite subset of $\Sigma$ is satisfiable. Completeness gives us a simpler proof of this theorem.

**Proof:** *(Sketch)* $\Sigma$ satisfiable $\iff$ $\Sigma$ consistent $\iff$ every finite subset of $\Sigma$ is consistent $\iff$ every finite subset of $\Sigma$ is satisfiable.

# 8 Introduction to Predicate Calculus (FOL)

Perhaps one of the most famous deductive examples is:

- Every man is mortal.
- Socrates is a man.
- Therefore, Socrates is mortal.

If we try to express this deduction in propositional logic we obtain three sentences with no relation to each other. Therefore, we get $\{P, Q\} \models R$, which is nonsensical. Propositional logic lacks the expressive power that demonstrates the relations between these three sentences.

Predicate Calculus (First-Order Logic) breaks a statement into two components: objects and properties. FOL is actually a large family of languages. Since FOL is a language, it relies on alphabets which consist of:

(a) **Variables** such as $v_0, \ldots, v_n, x, y$ that serve as placeholders for objects in the domain under consideration.

(b) **Logical Operators** such as $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$. However, minimally, we only consider $\neg$ and $\wedge$ since the others could all be derived using these two.

(c) **Logical quantifiers** $\exists$ and $\forall$ (Again, minimally, we only consider $\exists$) where quantification is restricted to objects only and not to formulae or sets of objects (but the objects themselves may be sets).

(d) **Equality Symbol** $=$

(e) **Constant Symbols** that stand for fixed individual objects in the domain.

(f) **Function Symbols** which stand for fixed functions that take objects as arguments and return objects as values. Each function as an associated arity (number of inputs it can take).

(g) **Relation Symbols** which stand for fixed relations between objects or their properties.

Fix a first-order language $\mathcal{L}$ consisting of:

- constant symbols $\{a_i\}_{i \in \mathbb{N}}$,
- variable symbols $\{x_i\}_{i \in \mathbb{N}}$,
- function symbols $\{f_i\}_{i \in \mathbb{N}}$ (each with specified arity),
- relation symbols $\{R_i\}_{i \in \mathbb{N}}$ (each with specified arity).

First, we define the collection of "words" that denote objects. We call these **terms**. The set of **terms** of $\mathcal{L}$ is defined inductively as follows:

1. Every variable and constant symbol is a term.
2. If $f$ is an $n$-ary function symbol and $t_1, \ldots, t_n$ are terms, then
$$f(t_1, \ldots, t_n)$$
is a term.
3. Nothing else is a term.

The set of **atomic formulas** consists of expressions of the form:

- $R(t_1, \ldots, t_n)$, where $R$ is an $n$-ary relation symbol and $t_1, \ldots, t_n$ are terms,
- $t_1 = t_2$, where $t_1, t_2$ are terms (if equality is included in the language).

The set of **well-formed formulas** (wffs) of $\mathcal{L}$ is defined inductively as the smallest set satisfying:

1. Every atomic formula is a wff.

2. If $\varphi$ and $\psi$ are wffs, then so are:

$$\neg\varphi, \quad (\varphi \wedge \psi), \quad (\varphi \vee \psi), \quad (\varphi \rightarrow \psi), \quad (\varphi \leftrightarrow \psi).$$

3. If $\varphi$ is a wff and $x$ is a variable, then:

$$\exists x\, \varphi \quad \text{and} \quad \forall x\, \varphi$$

   are wffs.

4. Nothing else is a wff.

Thus, the collection of wffs in a language $\mathcal{L}$ is generated inductively from atomic formulas using logical connectives and quantifiers.

**Free Variable:** Any variable that is not bounded by a quantifier is known as a free variable. $\mathbf{F}(\varphi)$ denotes the set of free variables in $\varphi$. We define this set inductively as well.

1. If $\varphi$ is an atomic formula:
   - If $\varphi \equiv R(t_1, \ldots, t_n)$, then
   $$\mathbf{F}(\varphi) = \bigcup_{i=1}^{n} \mathbf{F}(t_i),$$
   where $\mathbf{F}(t_i)$ denotes the variables occurring in the term $t_i$.
   - If $\varphi \equiv (t_1 = t_2)$, then
   $$\mathbf{F}(\varphi) = \mathbf{F}(t_1) \cup \mathbf{F}(t_2).$$

2. For terms, define $\mathbf{F}(t)$ inductively:
   - If $t$ is a variable $x$, then $\mathbf{F}(t) = \{x\}$.
   - If $t$ is a constant symbol, then $\mathbf{F}(t) = \varnothing$.
   - If $t \equiv f(t_1, \ldots, t_n)$, then
   $$\mathbf{F}(t) = \bigcup_{i=1}^{n} \mathbf{F}(t_i).$$

3. If $\varphi \equiv \neg\psi$, then
$$\mathbf{F}(\varphi) = \mathbf{F}(\psi).$$

4. If $\varphi \equiv (\psi \circ \theta)$, where $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then
$$\mathbf{F}(\varphi) = \mathbf{F}(\psi) \cup \mathbf{F}(\theta).$$

5. If $\varphi \equiv \exists x\, \psi$ or $\varphi \equiv \forall x\, \psi$, then
$$\mathbf{F}(\varphi) = \mathbf{F}(\psi) \setminus \{x\}.$$

**Remark:** A formula $\varphi$ is called a *sentence* if $\mathbf{F}(\varphi) = \varnothing$.

# 9 Semantics in FOL

Given a first-order language $\mathcal{L}$, an **$\mathcal{L}$-structure** (or *interpretation*) $\mathcal{M}$ consists of the following data:

- A non-empty set $M$, called the **universe** or **domain** of $\mathcal{M}$.
- For each constant symbol $c \in \mathcal{L}$, an element $c^{\mathcal{M}} \in M$.
- For each $n$-ary function symbol $f \in \mathcal{L}$, a function
$$f^{\mathcal{M}} : M^n \to M.$$
- For each $n$-ary relation symbol $R \in \mathcal{L}$, a relation
$$R^{\mathcal{M}} \subseteq M^n.$$
- If equality is part of the language, it is interpreted as actual equality on $M$, i.e.,
$$=^{\mathcal{M}} \; = \; \{(a, a) \mid a \in M\}.$$

**Remark:** We often write
$$\mathcal{M} = (M, (c^{\mathcal{M}})_{c \in \mathcal{L}}, (f^{\mathcal{M}})_{f \in \mathcal{L}}, (R^{\mathcal{M}})_{R \in \mathcal{L}})$$
to emphasize the interpretation of the symbols of $\mathcal{L}$.

Let $\mathcal{M}$ be an $\mathcal{L}$-structure with domain $M$, and let $V$ be the set of variables.

An **assignment function** is a map
$$s : V \to M,$$
which assigns to each variable an element of the domain.

We extend $s$ to a function
$$\bar{s} : T \to M,$$
where $T$ is the set of terms of $\mathcal{L}$, called the **evaluation of terms under** $s$. This extension is defined inductively as follows:

1. If $t$ is a variable $x$, then
$$\bar{s}(x) = s(x).$$
2. If $t$ is a constant symbol $c$, then
$$\bar{s}(c) = c^{\mathcal{M}}.$$
3. If $t \equiv f(t_1, \ldots, t_n)$, where $f$ is an $n$-ary function symbol, then
$$\bar{s}(f(t_1, \ldots, t_n)) \; = \; f^{\mathcal{M}}\big(\bar{s}(t_1), \ldots, \bar{s}(t_n)\big).$$

**Remark:** The function $\bar{s}$ is uniquely determined by $s$ and the structure $\mathcal{M}$.

We now extend the evaluation of terms to formulas by defining when a formula is **satisfied** in a structure under an assignment.

Let $\mathcal{M}$ be an $\mathcal{L}$-structure with domain $M$, and let $s : V \to M$ be an assignment. We define, inductively, what it means for a formula $\varphi$ to be satisfied in $\mathcal{M}$ under $s$, written

$$\mathcal{M} \models \varphi[s].$$

1. **Atomic formulas:**
   - If $\varphi \equiv R(t_1, \dots, t_n)$, then
   $$\mathcal{M} \models R(t_1, \dots, t_n)[s] \quad \text{iff} \quad \big(\bar{s}(t_1), \dots, \bar{s}(t_n)\big) \in R^{\mathcal{M}}.$$
   - If $\varphi \equiv (t_1 = t_2)$, then
   $$\mathcal{M} \models (t_1 = t_2)[s] \quad \text{iff} \quad \bar{s}(t_1) = \bar{s}(t_2).$$
2. **Negation:**
   $$\mathcal{M} \models \neg\psi[s] \quad \text{iff} \quad \mathcal{M} \not\models \psi[s].$$
3. **Conjunction:**
   $$\mathcal{M} \models (\psi \wedge \theta)[s] \quad \text{iff} \quad \mathcal{M} \models \psi[s] \text{ and } \mathcal{M} \models \theta[s].$$
4. **Other connectives:** Defined in the usual way from $\neg$ and $\wedge$.
5. **Existential quantifier:**
   $$\mathcal{M} \models \exists x\, \psi[s] \quad \text{iff} \quad \text{there exists } a \in M \text{ such that } \mathcal{M} \models \psi[s[x \mapsto a]].$$
6. **Universal quantifier:**
   $$\mathcal{M} \models \forall x\, \psi[s] \quad \text{iff} \quad \text{for all } a \in M,\ \mathcal{M} \models \psi[s[x \mapsto a]].$$

**Notation:** The assignment $s[x \mapsto a]$ denotes the function that agrees with $s$ on all variables except possibly $x$, and sends $x$ to $a$.

**Remark:** If $\varphi$ is a sentence (i.e., has no free variables), then its truth does not depend on the assignment $s$, and we simply write

$$\mathcal{M} \models \varphi.$$

**Theorem 9.1.** *Every wff has a unique decomposition.*

*Proof.* We prove this by induction on the construction of formulas. *Existence* follows directly from the inductive definition of wffs. For *uniqueness*, suppose a formula $\varphi$ admits two decompositions according to the formation rules. We argue by cases on the outermost form of $\varphi$:

- **Case 1: $\varphi$ begins with a propositional variable.** By definition, $\varphi$ must be atomic. Since no atomic formula is a prefix of another (or of a complex formula starting with symbols like '(' or '$\neg$'), the decomposition is unique.
- **Case 2: $\varphi$ begins with '$\neg$'.** Then $\varphi \equiv \neg\psi$ and $\varphi \equiv \neg\psi'$. This implies $\psi \equiv \psi'$. By the inductive hypothesis, $\psi$ has a unique decomposition.

- **Case 3: $\varphi$ begins with '('.** Then $\varphi \equiv (\psi \circ \theta)$ and $\varphi \equiv (\psi' \star \theta')$, where $\circ, \star$ are binary connectives. Suppose $\psi \not\equiv \psi'$. Then one must be a proper prefix of the other. Without loss of generality, let $\psi$ be a proper prefix of $\psi'$. Then we know that, $L(\psi) = R(\psi)$. However, since $\psi$ is a proper prefix of $\psi'$, and $\psi'$ is a wff, any proper prefix of $\psi'$ must have more left than right parentheses. This is a contradiction. Thus, $\psi \equiv \psi'$. It follows that $\circ$ must be the same symbol as $\star$, and consequently $\theta \equiv \theta'$.
- **Case 4: $\varphi$ begins with a quantifier.** Then $\varphi \equiv \forall x \psi$ and $\varphi \equiv \forall y \psi'$. The symbol following '$\forall$' must be the variable, so $x \equiv y$. This leaves $\psi \equiv \psi'$, which is unique by the inductive hypothesis.

In all cases, the outermost structure and the immediate subformulas are uniquely determined. By the induction hypothesis, the subformulas themselves have unique decompositions. Therefore, $\varphi$ has a unique decomposition. $\square$

**Corollary 9.1.** *$\bar{s}$ is well-defined.*

**An Example to bring everything together:** We define a first-order language for graph theory.

- A (possibly empty) set of constant symbols $\{c_i\}_{i \in I}$, which may be used to name specific vertices.
- A single binary relation symbol $E$, intended to represent the edge relation.
- No function symbols.

An $\mathcal{L}$-structure $\mathcal{M}$ for this language consists of:

- A non-empty set $V$, whose elements are interpreted as the vertices of the graph.
- An interpretation $E^{\mathcal{M}} \subseteq V \times V$, representing the edge relation.
- For each constant symbol $c_i$, an element $c_i^{\mathcal{M}} \in V$.

**Interpretation:** The pair $(V, E^{\mathcal{M}})$ can be viewed as a (directed) graph. If we wish to model undirected graphs, we may impose the additional condition:

$$(x, y) \in E^{\mathcal{M}} \iff (y, x) \in E^{\mathcal{M}}.$$

**Example Formula:** The formula

$$\forall x \, \forall y \, \big(E(x, y) \to E(y, x)\big)$$

expresses that the graph is undirected.

**Claim:** It $s_1$ and $s_2$ agree on the free variables of a term $t$, then $s_1(t) = s_2(t)$.

*Proof.* We prove this by structural induction on terms.

**Base Case:** Let $t$ be a variable. Then clearly it's a free variable, hence, $s_1(t) = s_2(t)$. The statement is vacuously true if $t$ is a constant.

**Inductive Step:** Let the property hold for $t_1, \ldots, t_n$ terms. Let $f$ be an $n$-ary function. Then,

$$s_1(f(t_1, \ldots, t_n)) = f^{\mathcal{M}}(s_1(t_1), \ldots, s_1(t_n)) \text{ and } s_2(f(t_1, \ldots, t_n)) = f^{\mathcal{M}}(s_2(t_1), \ldots, s_2(t_n))$$

Recall that by the inductive hypothesis $s_1(t_i) = s_2(t_i)$ for all $i \in 1, \ldots, n$. Hence, $s_1(f(t_1, \ldots, t_n)) = s_2(f(t_1, \ldots, t_n))$. $\square$

**Lemma 9.1.** *Let $\varphi$ be a wff in some language $\mathcal{L}$ with a structure $\mathcal{M}$. Then, given two assignments $s_1$ and $s_2$, if they agree on all free variables of $\varphi$, then $\mathcal{M} \models_{s_1} \varphi$ iff $\models_{s_2} \varphi$.*

*Proof.* We proceed with structural induction on $\varphi$.

**Base Case:** $\varphi$ is an atomic formula.

- $\varphi \equiv (t_1 = t_2)$: $\mathcal{M} \models_{s_1} \varphi$ iff $s_1(t_1) = s_1(t_2)$. By the earlier claim, we saw that $s_1(t) = s_2(t)$ for all terms $t$. Hence, $\mathcal{M} \models_{s_1} \varphi$ iff $s_2(t_1) = s_2(t_2)$ iff $\mathcal{M} \models_{s_2} \varphi$
- $\varphi \equiv R(t_1, \ldots, t_n)$: $\mathcal{M} \models_{s_1} \varphi$ iff $(s_1(t_1), \ldots, s_1(t_n)) \in R^{\mathcal{M}}$. Again, by the earlier claim, we saw that $s_1(t) = s_2(t)$ for all terms $t$. Hence, $\mathcal{M} \models_{s_1} \varphi$ iff $(s_2(t_1), \ldots, s_2(t_n)) \in R^{\mathcal{M}}$ iff $\mathcal{M} \models_{s_2} \varphi$.

**Inductive Hypothesis:** Let $\psi$ and $\phi$ satisfy the desired property.

- $\varphi \equiv \neg\psi$: Then $\mathbf{F}(\varphi) = \mathbf{F}(\psi)$. Hence, by the inductive hypothesis, $\mathcal{M} \models_{s_1} \varphi \iff \mathcal{M} \models_{s_2} \varphi$.
- $\varphi \equiv \psi \wedge \phi$: Then $(\mathcal{M} \models_{s_1} \varphi)$ iff $(\mathcal{M} \models_{s_1} \psi$ and $\mathcal{M} \models_{s_1} \phi)$ iff $(\mathcal{M} \models_{s_2} \psi$ and $\mathcal{M} \models_{s_2} \phi$ by induction hypothesis) iff $(\mathcal{M} \models_{s_2} \varphi)$.
- $\varphi \equiv \forall x\psi$: Then $\mathbf{F}(\varphi) = \mathbf{F}(\psi) \backslash \{x\}$. Then $\mathcal{M} \models_{s_1} \forall x\psi \iff$ for all $a \in M$, $\mathcal{M} \models_{s_1[x \to a]} \psi \iff$ for all $a \in M$ $\mathcal{M} \models_{s_2[x \to a]} \psi \iff \mathcal{M} \models_{s_2} \forall x\psi$ Hence, by the induction hypothesis, it follows that $\mathcal{M} \models_{s_1} \varphi$ iff $\mathcal{M} \models_{s_2} \varphi$.

  It's sufficient to demonstrate these three cases.

$\square$

Now, we can define a few semantic notions: 1. $\varphi$ is a **logical truth** if for every structure $\mathcal{M}$ and every assignment $s$, $\mathcal{M} \models_s \varphi$.

**Examples:** Let $\varphi \equiv \forall x Q(x) \to Q(y)$ where $Q$ is a relation with arity 1. Assume that for a given $\mathcal{M}, s$, $\mathcal{M} \not\models_s \varphi$. Then, by truth table, $\mathcal{M} \not\models_s Q(y)$. Hence, $\bar{s}(y)$ does not satisfy $Q^{\mathcal{M}}$. But if that's the case then $\mathcal{M} \not\models_s \forall x Q(x)$ since $\bar{s}(y) \in M$ doesn't satisfy $Q^{\mathcal{M}}$. Hence $\mathcal{M} \models_s \varphi$.

**Example:** $\exists x(P(x) \to \forall x P(x))$ is a logical truth (why?).

2. $\Sigma$ **logically implies** $\varphi$ if for every $\mathcal{M}, s$ that make every element of $\Sigma$ true, $\mathcal{M} \models_s \varphi$.

# 10 Expressive Power of FOL

A few clarification on notations:

- $\mathcal{M} \models_s \varphi$: The formula $\varphi$ gets value $T$ in the structure $\mathcal{M}$ under the assignment $s$.
- $s : V \to M$ is the assignment mapping variables to $M$. $\bar{s}$ is the extension of $s$ to terms.

**Definition 10.1.** *A set of structures $K$ is definable in a first-order language $\mathcal{L}$, if there exists a set of $\mathcal{L}$-sentences $\Sigma$ such that $K = \{\mathcal{M} : \mathcal{M} \models \Sigma\}$.*

Let $\mathcal{L}$ be the empty language. We claim the following:

1. The collection $K$ of all structures $\mathcal{M}$ such that $|M| = 1$ is definable in $\mathcal{L}$. Define $\Sigma_1 = \{\forall x \forall y(x = y)\}$. Then clearly for every $\mathcal{M} \in K$, $\mathcal{M} \models \Sigma_1$.

2. For every $n$, the collection $K$ of all structures $\mathcal{M}$ with at least $n$ elements in $M$ is definable. Define $\Sigma = \{\exists x_1 \exists x_2 \dots \exists x_n \bigwedge_{i \neq j} \neg(x_i = x_j)\}$. Then every structure in $K$ satisfies $\Sigma$.

3. For every $n$, the collection $K$ of all structures $\mathcal{M}$ with at most $n$ elements in $M$ is definable. Define $\Sigma = \{\forall x_1 \dots \forall x_n \forall x_{n+1} \bigvee_{i \neq j}(x_i = x_j)\}$. Then every structure in $K$ satisfies $\Sigma$.

4. The collection $K$ of all structures $\mathcal{M}$ with infinite elements in $M$ is definable. Let $\varphi_n = \exists x_1 \exists x_2 \dots \exists x_n \bigwedge_{i \neq j} \neg(x_i = x_j)$. Define $\Sigma = \{\varphi_n\}_{n \in \mathbb{N}}$.

Let our language $\mathcal{L}$ now consist of a single relation $R$ of arity 2. We define the set $K$ of all structures $\mathcal{M}$ such that $R^{\mathcal{M}}$is a total order. Then we have $\Sigma = \{\forall x, \forall y(R(x, y) \vee R(y, x)), \forall x \forall y \forall z(R(x, y) \wedge R(y, z) \rightarrow R(x, z)), \forall x \forall y(R(x, y) \wedge R(y, x) \rightarrow x = y)\}$. This set makes $K$ definable.

**Definability Within a structure:** Let $\mathcal{M}$ be a structure of a language $\mathcal{L}$ given by:

$$\mathcal{M} = \langle \mathbb{N}, +, \cdot, i \rangle$$

Where $i$ is the successor function. Now, we can choose subsets of $\mathcal{M}$ that are definable. This is what definability within a structure means. A statement $\varphi(x)$ defines a subset $A \subseteq M$ if $\mathcal{M} \models_s \varphi(x) \iff s(x) \in A$.

# 11 Applications of the Compactness Theorem

We'll state the compactness theorem without proof for now. We'll return to the prove once we have established soundness and completeness.

**Theorem 11.1** (Compactness Theorem)**.** *A set of wff is satisfiable if and only if every finite subset of it is satisfiable.*

We can now explore some interesting applications of compactness theorem.

Recall that a family of structures $K$ is definable if there exists a set of wffs $\Sigma$ such that $K = \{\mathcal{M} : \mathcal{M} \models \Sigma\}$.

Given the set $K$ which sis the family of all finite structures, we make the following claim.

**Claim:** $K$ is not definable.

*Proof.* Let $\Gamma = \{\varphi_n : n \in \mathbb{N}\}$ where $\varphi_n \equiv \exists x_1 \exists x_2 \dots \exists x_n \left( \bigwedge_{i \neq j,\, i,j \leqslant n} \neg(x_i = x_j) \right)$, so that any structure satisfying $\Gamma$ must be infinite.

Suppose for contradiction that $K$ is definable, i.e., there exists a set of wffs $\Sigma$ whose models are exactly the finite structures. Consider the set $\Sigma \cup \Gamma$. We claim every finite subset of $\Sigma \cup \Gamma$ is satisfiable. Let $\Delta$ be any finite subset of $\Sigma \cup \Gamma$. Then $\Delta \cap \Gamma$ is finite, so it contains only finitely many sentences $\varphi_n$; let $N$ be the largest such $n$. Then any structure with exactly $N$ elements satisfies every sentence in $\Delta \cap \Gamma$, and since such a structure is finite, it also satisfies every sentence in $\Delta \cap \Sigma$ (as $\Sigma$ is satisfied by all finite structures). Hence $\Delta$ is satisfiable.

Since every finite subset of $\Sigma \cup \Gamma$ is satisfiable, Compactness gives us a model $\mathcal{M} \models \Sigma \cup \Gamma$. Since $\mathcal{M} \models \Sigma$, and the models of $\Sigma$ are exactly the finite structures, $\mathcal{M}$ must be finite. But since $\mathcal{M} \models \Gamma$, $\mathcal{M}$ must be infinite. This is a contradiction. Therefore no such $\Sigma$ exists, and $K$ is not definable. $\square$

Now, consider the family of graphs $G = (V, E)$. A graph has a diameter $n$ if you can go from some vertex to another vertex by at most $n$ steps.

Say that the diameter of $G$ is k if for every $u, v \in V$ there is a path $uw_1 \dots w_{k-1}v$ such that $(u, w_1), (w_i, w_{i+1}), (w_k, v) \in E$ for all $i \leqslant k$. Can we find a $\Gamma$ that defines the set of all graphs with infinite diameter. We define $\Gamma$ as:

$$\Gamma = \{\varphi_k : k \in \mathbb{N}\}$$

$\varphi_1 \equiv \exists x \exists y \neg R(x, y) \wedge \neg R(x, y)$ and $\varphi_k \equiv \exists x \exists y(\neg \exists v_1, \dots, v_{k-1}(R(x, v_1) \wedge R(v_1, v_2) \wedge \dots \wedge R(v_{k-1}, y))\}$

**Claim:** The family of all finite diameter graphs is not definable.

*Proof.* Assume $\Sigma$ defines such a family. We now define a union $\Sigma \cup \Gamma$, where $\Gamma$ defines the family of all infinite diameter graphs. Let a finite $\Delta \subseteq \Sigma \cup \Gamma$. We know that $\Delta = (\Delta \cap \Sigma) \cup (\Delta \cap \Gamma)$. Any $\Delta \cap \Gamma$ is finite with a maximum $k$. Then any graph with diameter $k + 1$ satisfies $\Delta \cap \Gamma$. Since a graph of diameter $k + 1$ has finite diameter, it is also a model of $\Sigma$, and hence satisfies $\Delta \cap \Sigma$. Hence the family of finite diameter graphs satisfy $\Delta$. Since any finite subset of $\Sigma \cup \Gamma$ is satisfiable, compactness tell us that there must exist a model $\mathcal{M}$ such that $\mathcal{M} \models \Sigma \cup \Gamma$. But $\mathcal{M} \models \Sigma$ implies $\mathcal{M}$ has finite diameter and $\mathcal{M} \models \Gamma$ implies $\mathcal{M}$ has infinite diameter. Hence, a contradiction. □

Now we see a more positive application of Compactness by exploring a non-standard model of arithmetic. Consider the language for number theory:

$$\mathcal{L} = \{f, g, i, R, a, b\}$$

Interpreted to a structure as:

$$\mathcal{M} = \{+, \cdot, s, , \leqslant, 0, 1\}$$

We now create the set of theories of the structure as:

$$\mathsf{Th}(\mathcal{M}) = \{\varphi : \varphi \text{ is a sentence in } \mathcal{L} \text{ and } \mathcal{M} \models \varphi\}$$

There exists a structure $\mathcal{N}$ that satisfies every sentence in $\mathsf{Th}(\mathcal{M})$ but as an element with infinitely many predecessors. Such an $\mathcal{N}$ is called a non-standard model of arithmetic.

**Theorem 11.2.** *There exists a non-standard model $\mathcal{N}$ of arithmetic exists.*

*Proof.* Let $c$ be a new constant symbol not in $\mathcal{L}$, and let $\mathcal{L}' = \mathcal{L} \cup \{c\}$. Define the set of sentences:

$$\Gamma = \{\varphi : \varphi \in \mathsf{Th}(\mathcal{M})\} \cup \{s^n(0) < c : n \in \mathbb{N}\}$$

where $s^n(0)$ denotes 0 with the successor function applied $n$ times, i.e., $\underbrace{s(s(\dots s}_{n}(0)\dots))$.

We claim every finite subset $\Delta \subseteq \Gamma$ is satisfiable. Let $\Delta$ be any finite subset. Then $\Delta$ contains only finitely many sentences of the form $s^n(0) < c$; let $N$ be the largest such $n$. Interpret $c$ as $N + 1$ in $\mathcal{M}$. Then $\mathcal{M}$ satisfies every sentence in $\Delta \cap \mathsf{Th}(\mathcal{M})$ by definition, and satisfies every sentence $s^n(0) < c$ in $\Delta$ since $n \leqslant N < N + 1$. Hence $\Delta$ is satisfiable.

Since every finite subset of $\Gamma$ is satisfiable, Compactness gives a model $\mathcal{N} \models \Gamma$. Since $\mathcal{N} \models \mathsf{Th}(\mathcal{M})$, the structure $\mathcal{N}$ satisfies every sentence true in $\mathcal{M}$. Moreover, $\mathcal{N}$ contains an element $c^{\mathcal{N}}$ satisfying $s^n(0) < c^{\mathcal{N}}$ for every $n \in \mathbb{N}$, meaning $c^{\mathcal{N}}$ has infinitely many predecessors. Since no standard natural number has infinitely many predecessors, $c^{\mathcal{N}}$ is not a standard natural number, and hence $\mathcal{N}$ is a non-standard model of arithmetic. □

# 12 Proof Systems in FOL

Let $\Delta$ be a proof system (Axioms + Deduction Rules) such that:

**1. Soundness:** For every set of wffs $\Sigma$ and every $\varphi$, if $\Sigma \vdash_\Delta \varphi$ then $\Sigma \models \varphi$.

**2. Completeness:** For every $\Sigma$ and $\varphi$, $\Sigma \models \varphi$ implies $\Sigma \vdash \varphi$

**3.** Every proof in the system $\Delta$ is finite.

Then this will be enough to prove the compactness theorem.

**Claim:** Such a $\Delta$ implies the compactness theorem.

*Proof.* Suppose $\Sigma$ is a set of wffs such that every finite subset of $\Sigma$ is satisfiable. We wish to show $\Sigma$ is satisfiable.

Suppose for contradiction that $\Sigma$ is unsatisfiable, i.e., $\Sigma \models \varphi$ for every sentence $\varphi$ (in particular, $\Sigma \models \varphi \wedge \neg\varphi$ for some $\varphi$). By Completeness, $\Sigma \vdash_\Delta \varphi \wedge \neg\varphi$.

Since every proof in $\Delta$ is finite, this proof uses only finitely many hypotheses from $\Sigma$. That is, there exists a finite subset $\Delta_0 \subseteq \Sigma$ such that $\Delta_0 \vdash_\Delta \varphi \wedge \neg\varphi$. By Soundness, $\Delta_0 \models \varphi \wedge \neg\varphi$, meaning $\Delta_0$ is unsatisfiable. But $\Delta_0$ is a finite subset of $\Sigma$, contradicting our assumption that every finite subset of $\Sigma$ is satisfiable.

Therefore $\Sigma$ is satisfiable, and Compactness holds. $\square$

We now define a proof system for first order logic.

Let $\varphi, \varphi_1, \varphi_2$, and $\psi$ be arbitrary first-order formulae:

$$\begin{aligned}
&\mathsf{L0}: && \varphi \vee \neg\varphi \\
&\mathsf{L1}: && \varphi \to (\psi \to \varphi) \\
&\mathsf{L2}: && (\psi \to (\varphi_1 \to \varphi_2)) \to ((\psi \to \varphi_1) \to (\psi \to \varphi_2)) \\
&\mathsf{L3}: && (\varphi \wedge \psi) \to \varphi \\
&\mathsf{L4}: && (\varphi \wedge \psi) \to \psi \\
&\mathsf{L5}: && \varphi \to (\psi \to (\psi \wedge \varphi)) \\
&\mathsf{L6}: && \varphi \to (\varphi \vee \psi) \\
&\mathsf{L7}: && \psi \to (\varphi \vee \psi) \\
&\mathsf{L8}: && (\varphi_1 \to \varphi_3) \to ((\varphi_2 \to \varphi_3) \to ((\varphi_1 \vee \varphi_2) \to \varphi_3)) \\
&\mathsf{L9}: && \neg\varphi \to (\varphi \to \psi)
\end{aligned}$$

If $t$ is a term, $x$ is a variable, and the substitution which leads to $\varphi[x \mapsto t]$ is admissible, then:

$$\begin{aligned}
&\mathsf{L10}: \forall x \varphi(x) \to \varphi[x \mapsto t] \\
&\mathsf{L11}: \varphi[x \mapsto t] \to \exists x \varphi(x)
\end{aligned}$$

If $\psi$ is a formula and $x$ is a variable such that $x \notin \text{free}(\psi)$ then:

$$\mathsf{L12} : \forall x(\psi \to \varphi(x)) \to (\psi \to \forall x\varphi(x))$$
$$\mathsf{L13} : \forall x(\varphi(x) \to \psi) \to (\exists x\varphi(x) \to \psi)$$

What's left is the symbol $=$. In our formal system, the binary equality relation is defined by the following three axioms.

If $t_1, \ldots, t_n, t'_1, \ldots, t'_n$ are any terms, $R$ is an $n$-ary relation symbol (for example, $=$) and $F$ is an $n$-ary function symbol, then:

$$\mathsf{L14} : t = t \tag{1}$$
$$\mathsf{L15} : (t_1 = t'_1 \wedge \cdots \wedge t_n = t'_n) \to (R(t_1, \ldots, t_n) \to R(t'_1, \ldots, t'_n)) \tag{2}$$
$$\mathsf{L16} : (t_1 = t'_1 \wedge \cdots \wedge t_n = t'_n) \to (F(t_1, \ldots, t_n) \to F(t'_1, \ldots, t'_n)) \tag{3}$$
$$\tag{4}$$

Finally, notice that $\tau_1 \neq \tau_2 \equiv \neg(\tau_1 = \tau_2)$ and $\psi \leftrightarrow \varphi \equiv (\psi \to \varphi) \wedge (\varphi \to \psi)$.

| Group | Axioms | What they do |
|---|---|---|
| Propositional | L0–L9 | All laws of $\neg$, $\wedge$, $\vee$, $\to$ |
| Instantiation | L10–L11 | Pull $\forall$ out / push $\exists$ in (with admissible substitution) |
| Quantifier shift | L12–L13 | Move $\forall/\exists$ past $\to$ when the other part is independent of the variable |
| Equality Axioms | L14-L16 | Define the behavior of $=$ |

**Example:** Prove that $\vdash P(x) \to \exists x P(x)$

**Proof:** Simple instantiation of $\mathsf{L11}$.

**Definition 12.1.** *For a set of formulas $\Gamma$ and a formula $\alpha$, we write $\Gamma \vdash \alpha$ iff $\alpha$ belongs to the inductive set generated by the Axioms (L1-L16), $\Gamma$, and Modus Ponens.*

**Theorem 12.1** (Soundness Theorem)**.** *For every set of formulas $\Gamma$ and formula $\alpha$, if $\Gamma \vdash \alpha$, then $\Gamma \models \alpha$.*

*Proof.* We proceed by induction on the length of the derivation of $\alpha$ from $\Gamma$.

**Base case.** If the derivation has length 1, then $\alpha$ is either a member of $\Gamma$ or an instance of an axiom schema (L0–L16). In the first case, any model satisfying every formula in $\Gamma$ satisfies $\alpha$ immediately. In the second case, one verifies by direct semantic inspection that every axiom schema is universally valid; for instance, L0 holds because every interpretation makes $\varphi \vee \neg\varphi$ true, L10 holds because if $\forall x\, \varphi(x)$ is true then $\varphi[x \mapsto t]$ is true for any admissibly substituted term $t$, and so on for the remaining schemas.[1]

**Inductive step.** Assume the claim holds for all derivations of length less than $n$, and suppose $\alpha$ is obtained at step $n$ by Modus Ponens from earlier lines $\beta$ and $\beta \to \alpha$. By the induction hypothesis,

[1] Each axiom schema is a logical truth by direct semantic inspection: L0 holds since every interpretation satisfies $\varphi \vee \neg\varphi$; L1–L2 hold since the consequent is true whenever the antecedent is; L3–L5 hold by the semantics of $\wedge$; L6–L8 hold by the semantics of $\vee$; L9 holds since a false formula implies anything; L10–L11 hold since $\forall x\, \varphi(x)$ implies $\varphi[x \mapsto t]$ and $\varphi[x \mapsto t]$ implies $\exists x\, \varphi(x)$ for any admissible term $t$; L12–L13 hold since when $x \notin \text{free}(\psi)$ the quantifier may be shifted past $\to$ without affecting the truth value of $\psi$; and L14–L16 hold since equality is interpreted as the identity relation, making reflexivity, substitutivity for relations, and substitutivity for functions all universally valid.

$\Gamma \models \beta$ and $\Gamma \models (\beta \to \alpha)$. Let $\mathcal{M}$ be any model and $s$ any assignment satisfying every formula in $\Gamma$. Then $\mathcal{M}, s \models \beta$ and $\mathcal{M}, s \models \beta \to \alpha$, so by the semantics of $\to$ we get $\mathcal{M}, s \models \alpha$. Since $\mathcal{M}$ and $s$ were arbitrary, $\Gamma \models \alpha$.

By induction, $\Gamma \vdash \alpha$ implies $\Gamma \models \alpha$. $\square$

Well state the following theorems on formal proofs (without proofs):

**1. Deduction Theorem:** For every $\Gamma, \alpha, \beta$, $\Gamma \cup \{\alpha\} \vdash \beta$ iff $\Gamma \vdash (\alpha \to \beta)$.

**2.** The following are equivalent:

1. There exists some $\beta$ such that $\Gamma \nvdash \beta$.
2. There is no $\alpha$ such that $\Gamma \vdash \alpha$ and $\Gamma \vdash \neg\alpha$.
3. $\Gamma$ is consistent

**Theorem 12.2** (Generalization Theorem)**.** *If $\Gamma \vdash \alpha$ and $x$ is not free in any formula in $\Gamma$, then $\Gamma \vdash \forall x \alpha$.*

*Proof.* If $x$ is not free in $\alpha$, then $\forall x\, \alpha \leftrightarrow \alpha$ semantically, and $\Gamma \vdash \forall x\, \alpha$ follows trivially from $\Gamma \vdash \alpha$.

Since $\Gamma \vdash \alpha$, pick any $\psi \in \Gamma$. Because $x \notin \text{free}(\psi)$, we can write the derivation as:

| Step | Formula | Justification |
|---|---|---|
| 1 | $\alpha$ | Given, $\Gamma \vdash \alpha$ |
| 2 | $\psi \to \alpha$ | From (1) by L1 |
| 3 | $\forall x(\psi \to \alpha)$ | From (2), $x \notin \text{free}(\psi)$, by L12 |
| 4 | $\forall x(\psi \to \alpha) \to (\psi \to \forall x\, \alpha)$ | L12 |
| 5 | $\psi \to \forall x\, \alpha$ | MP on (3) and (4) |
| 6 | $\forall x\, \alpha$ | MP on $\psi$ and (5) |

Formally, since $\Gamma \vdash \alpha$ and $x \notin \text{free}(\Gamma)$, the rule encoded by L12 allows us to universally close $\alpha$ over $x$, yielding $\Gamma \vdash \forall x\, \alpha$. $\square$

We state the Completeness theorem without proof.

**Theorem 12.3** (Completeness)**.** *If $\Gamma \models \alpha$, then $\Gamma \vdash \alpha$.*

**Theorem 12.4.** *$\Gamma$ is consistent iff $\Gamma$ is satisfiable.*

*Proof.* ($\Rightarrow$) Assume $\Gamma$ is consistent. Let no $\mathcal{M}$ exists such that $\mathcal{M} \models \Gamma$. Then $\Gamma \models \beta$ for any $\beta$. By completeness, it follows that $\Gamma \vdash \beta$ for all $\beta$, which contradicts the consistency of $\Gamma$.

($\Leftarrow$) Let $\Gamma$ be satisfiable. Then there exists $\mathcal{M}$ such that $\mathcal{M} \models \Gamma$. Suppose for contradiction that $\Gamma$ is inconsistent, i.e. $\Gamma \vdash \beta \wedge \neg\beta$ for some $\beta$. By soundness, $\Gamma \models \beta \wedge \neg\beta$. But then $\mathcal{M} \models \beta \wedge \neg\beta$, which is a contradiction since no interpretation can satisfy both $\beta$ and $\neg\beta$ simultaneously. Hence $\Gamma$ is consistent. $\square$

# 13 What We've Done So Far

| **Syntax** | **Semantics** | **Bridging Theorems / Connections** |
|---|---|---|
| | *Basic Building Blocks* | |
| Terms (variables, constants, function applications) | Assignment function $s : V \to M$ and its extension $\bar{s} : T \to M$ | **Claim (Term Coincidence):** If $s_1, s_2$ agree on all free variables of $t$, then $\bar{s}_1(t) = \bar{s}_2(t)$. Syntax (free variables) determines which part of the assignment matters. |
| Well-formed formulas (wffs), built inductively from atomic formulas, connectives, and quantifiers | Satisfaction relation $\mathcal{M} \models_s \varphi$, defined inductively over structures and assignments | **Unique Decomposition Theorem:** Every wff has a unique syntactic decomposition, ensuring $\bar{s}$ is well-defined and the semantic interpretation is unambiguous. |
| Free variables $\mathbf{F}(\varphi)$, defined syntactically by induction on $\varphi$ | Truth value of $\varphi$ in $\mathcal{M}$ under assignment $s$ | **Coincidence Lemma:** $\mathcal{M} \models_{s_1} \varphi \iff \mathcal{M} \models_{s_2} \varphi$ whenever $s_1, s_2$ agree on $\mathbf{F}(\varphi)$. Only syntactically free variables affect truth. |
| Sentences (wffs with $\mathbf{F}(\varphi) = \varnothing$) | Truth in a structure $\mathcal{M} \models \varphi$ (no assignment needed) | Sentences have assignment-independent truth values; syntax (no free variables) licenses a purely semantic statement. |
| | *Proof Theory vs. Model Theory* | |
| Formal derivability $\Gamma \vdash \varphi$ (syntactic deduction via axioms L0–L16 and Modus Ponens) | Logical consequence $\Gamma \models \varphi$ (truth in every model satisfying $\Gamma$) | **Soundness Theorem:** $\Gamma \vdash \varphi \Rightarrow \Gamma \models \varphi$. Every syntactic proof is semantically valid. **Completeness Theorem (Gödel):** $\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi$. Every semantic truth has a syntactic proof. Together, $\vdash$ and $\models$ coincide. |
| Consistency of $\Gamma$ (no $\alpha$ with $\Gamma \vdash \alpha$ and $\Gamma \vdash \neg\alpha$) | Satisfiability of $\Gamma$ (existence of a model $\mathcal{M} \models \Gamma$) | **Consistency $\iff$ Satisfiability:** Syntactic consistency is equivalent to semantic satisfiability, proved using Soundness ($\Rightarrow$) and Completeness ($\Leftarrow$). |
| Finite proofs (every derivation uses finitely many steps and hypotheses) | Satisfiability of arbitrary (possibly infinite) sets of sentences | **Compactness Theorem:** $\Gamma$ is satisfiable $\iff$ every finite $\Delta \subseteq \Gamma$ is satisfiable. Follows from Soundness + Completeness + finiteness of proofs. |
| | *Expressive Power* | |
| | | *Continued on next page...* |

| Syntax | Semantics | Bridging Theorems / Connections |
|---|---|---|
| Set of $\mathcal{L}$-sentences $\Sigma$ (syntactic object) | Definable class of structures $K = \{\mathcal{M} : \mathcal{M} \models \Sigma\}$ (semantic object) | **Definability:** A class $K$ is definable iff there is a $\Sigma$ whose models are exactly $K$. Compactness limits expressibility; e.g., finite structures and finite-diameter graphs are *not* definable. |
| A formula $\varphi(x)$ with one free variable (syntactic) | Definable subset $A = \{a \in M : \mathcal{M} \models_{s[x \mapsto a]} \varphi\}$ of the domain (semantic) | **Definability within a structure:** Syntax (formulas with free variables) carves out semantic subsets of the domain. |
| | *Quantifier Reasoning* | |
| Generalization rule: if $\Gamma \vdash \alpha$ and $x \notin \text{free}(\Gamma)$, derive $\Gamma \vdash \forall x\, \alpha$ | Universal truth over the domain: $\mathcal{M} \models \forall x\, \alpha$ for all $a \in M$ | **Generalization Theorem:** The syntactic condition ($x$ not free in $\Gamma$) guarantees the semantic universal generalisation is valid. |
| Axioms L10–L13 governing $\forall$ and $\exists$ (syntactic rules) | Semantics of quantifiers: ranging over all $a \in M$ in $\mathcal{M} \models \forall x\, \psi[s]$ | L10–L13 are sound (Soundness Theorem): every syntactic quantifier manipulation corresponds to a valid semantic move. |

Table 1: The syntax–semantics correspondence in First-Order Logic, with bridging theorems.

## 14 Gödel's Incompleteness Theorem

The Soundness and Completeness Theorems together establish that the proof system $\Delta$ (axioms L0–L16 with Modus Ponens) is a *perfect* match for first-order logical consequence: $\Gamma \vdash \varphi$ if and only if $\Gamma \models \varphi$. One might hope that for a specific, sufficiently rich theory $T$, such as arithmetic, completeness means $T$ can decide every sentence, i.e., for every sentence $\varphi$, either $T \vdash \varphi$ or $T \vdash \neg\varphi$. Gödel's Incompleteness Theorems shatter this hope.

Hilbert, one of the greatest mathematicians of his time, attempted what's known as "Hibert's Program", which aimed to formalize the notion of a valid mathematical proof. A proof should satisfy three requirements:

1. **Soundness:** for every first-order statement on the natural numbers $\varphi$, if $\vdash \varphi$, then $\varphi$ is a correct statement.
2. **Completeness:** every true first-order statement $\varphi$ on the natural systems is provable.
3. **Verifiability:** there should be a clear formal method to check if a given string of symbols is a valid proof.

Looking at this requirements, it seems that FOL proof system is indeed verifiable, complete, and sound with respect to $\{\varphi : \varphi \text{ is a logical truth}\}$. But note that this set is smaller than the set of first-order statements on the natural numbers, i.e., there are statements that are true to the natural numbers but fail under another structure.

Peano Arithmetic is the standard proof system we use when working on $\mathbb{N}$. This system satisfies Soundness and Verifiability. And it was hoped that it would be complete. This is where Godel steps in.

**Godel's Incompleteness:** For every proof system in number theory that is verifiable, there is a statement $\varphi$ about numbers such that $\nvdash \varphi$ and $\nvdash \neg\varphi$.

This statement holds for any rich enough mathematical object.

## Background and Setup

Recall from the section on non-standard models the language of arithmetic:

$$\mathcal{L} = \{+, \cdot, s, \leqslant, 0, 1\}$$

interpreted in the standard model $\mathcal{M} = (\mathbb{N}, +, \cdot, s, \leqslant, 0, 1)$. We defined the **theory of arithmetic**:

$$\mathsf{Th}(\mathcal{M}) = \{\varphi : \varphi \text{ is an } \mathcal{L}\text{-sentence and } \mathcal{M} \models \varphi\}.$$

We now work with a weaker but more tractable theory. Let $\mathsf{PA}$ (*Peano Arithmetic*) denote a standard recursive axiomatisation of arithmetic. We will not list all axioms of $\mathsf{PA}$ here, but its key properties are:

- $\mathsf{PA}$ is **consistent**: no sentence $\varphi$ satisfies both $\mathsf{PA} \vdash \varphi$ and $\mathsf{PA} \vdash \neg\varphi$. By our Consistency $\iff$ Satisfiability theorem, $\mathsf{PA}$ has a model.
- $\mathsf{PA}$ is **recursively axiomatisable**: there is an algorithm that decides whether any given sentence is an axiom of $\mathsf{PA}$.
- $\mathsf{PA}$ is **sufficiently strong**: it can represent all primitive recursive functions and relations, and in particular can encode and reason about finite syntactic objects (formulas, proofs) as natural numbers.

The third point is crucial and deserves elaboration.

## Gödel Numbering (Sketch)

The central technical idea is that every syntactic object in $\mathcal{L}$ — every symbol, term, formula, and finite sequence of formulas (i.e., proof) — can be assigned a unique natural number called its **Gödel number**, written $\ulcorner\varphi\urcorner$. This encoding is entirely explicit and computable. Its key consequence is:

- The property "$n$ is the Gödel number of a formula" is decidable by $\mathsf{PA}$.
- The property "$n$ is the Gödel number of a proof of the formula with Gödel number $m$" is decidable by $\mathsf{PA}$.
- Consequently, the provability predicate $\mathsf{Prov}(m) \equiv \exists n\, \mathsf{Proof}(n, m)$ ("there exists a proof of the formula numbered $m$") is expressible as an $\mathcal{L}$-formula.

This means that $\mathsf{PA}$ can *talk about its own proofs* inside the object language. We now have the tools to state and prove the main results.

## The Diagonal Lemma

The following lemma is the engine behind both incompleteness theorems. Its proof uses the Gödel numbering machinery in full.

**Lemma 14.1** (Diagonal Lemma)**.** *Let $T$ be any consistent, recursively axiomatisable extension of* $\mathsf{PA}$*, and let $\psi(x)$ be any $\mathcal{L}$-formula with one free variable. Then there exists a sentence $\varphi$ such that:*

$$T \vdash \varphi \leftrightarrow \psi(\ulcorner \varphi \urcorner).$$

*That is, $\varphi$ "says of itself" that it satisfies $\psi$.*

*Proof (Sketch).* Define the *substitution function* $\mathsf{sub}(m,n)$ to be the Gödel number of the formula obtained by substituting the numeral $\bar{n}$ for the free variable in the formula numbered $m$. This function is representable in $\mathsf{PA}$.

Now let $\theta(x) \equiv \psi(\mathsf{sub}(x,x))$, and set $k = \ulcorner \theta \urcorner$. Define $\varphi \equiv \theta(\bar{k}) \equiv \psi(\mathsf{sub}(\bar{k},\bar{k}))$. Then $\mathsf{sub}(k,k) = \ulcorner \varphi \urcorner$ by construction, and $T \vdash \varphi \leftrightarrow \psi(\ulcorner \varphi \urcorner)$. □

## First Incompleteness Theorem

**Theorem 14.1** (Gödel's First Incompleteness Theorem)**.** *Let $T$ be a consistent, recursively axiomatisable theory that extends* $\mathsf{PA}$*. Then $T$ is* ***incomplete****: there exists a sentence $G$ such that*

$$T \nvdash G \quad \textit{and} \quad T \nvdash \neg G.$$

*Proof.* Let $\mathsf{Prov}_T(x)$ be the provability predicate for $T$, i.e., the $\mathcal{L}$-formula expressing "the formula with Gödel number $x$ is provable in $T$". Apply the Diagonal Lemma to $\psi(x) \equiv \neg\mathsf{Prov}_T(x)$ to obtain a sentence $G$ such that:

$$T \vdash G \leftrightarrow \neg\mathsf{Prov}_T(\ulcorner G \urcorner). \tag{$*$}$$

Informally, $G$ says: *"I am not provable in $T$."*

We now show $T \nvdash G$ and $T \nvdash \neg G$.

**Case 1: $T \nvdash G$.**

Suppose for contradiction that $T \vdash G$. Then $\mathsf{Prov}_T(\ulcorner G \urcorner)$ is true (there genuinely exists a proof), and since $T$ is sufficiently strong, $T \vdash \mathsf{Prov}_T(\ulcorner G \urcorner)$. But by $(*)$, $T \vdash G \leftrightarrow \neg\mathsf{Prov}_T(\ulcorner G \urcorner)$, so $T \vdash \neg G$. This contradicts the consistency of $T$.

**Case 2: $T \nvdash \neg G$.**

Suppose for contradiction that $T \vdash \neg G$. By $(*)$, $T \vdash \mathsf{Prov}_T(\ulcorner G \urcorner)$, meaning $T$ proves that $G$ is provable. But we showed in Case 1 that $G$ is in fact not provable. So $T$ proves a false arithmetic statement about provability. More precisely, $T \vdash \mathsf{Prov}_T(\ulcorner G \urcorner)$, yet no proof of $G$ exists, so this sentence is false in $\mathcal{M}$, meaning $T$ has no model in which $\mathcal{M}$ is the standard natural numbers.[2]

Therefore, neither $G$ nor $\neg G$ is provable in $T$, so $T$ is incomplete. □

[2] To conclude $T \nvdash \neg G$ without appealing to the standard model, one assumes the stronger hypothesis that $T$ is $\omega$-*consistent*: $T$ does not prove $\exists x\, \neg\psi(x)$ when it proves $\neg\psi(\bar{n})$ for every standard numeral $\bar{n}$. Under $\omega$-consistency, the argument is purely syntactic. Rosser's 1936 refinement removes even this assumption, requiring only plain consistency.

**Remark:** The sentence $G$ is *true* in the standard model $\mathcal{M}$: since $T \nvdash G$, the statement "$G$ is not provable" is correct. This shows that $\mathsf{Th}(\mathcal{M})$ strictly contains $T$ — there are true arithmetic sentences that $T$ cannot prove. Recall that we constructed a non-standard model $\mathcal{N} \models \mathsf{Th}(\mathcal{M})$ using Compactness; that model also satisfies $G$. Since $T \nvdash G$, by Completeness there must exist *some* model of $T$ in which $G$ fails — and indeed, non-standard models of $T$ alone (not of the full $\mathsf{Th}(\mathcal{M})$) can satisfy $\neg G$.

## Second Incompleteness Theorem

**Theorem 14.2** (Gödel's Second Incompleteness Theorem)**.** *Let $T$ be a consistent, recursively axiomatisable extension of* PA*. Then:*

$$T \nvdash \mathsf{Con}(T),$$

*where* $\mathsf{Con}(T)$ *is the $\mathcal{L}$-sentence asserting the consistency of $T$, i.e.,* $\neg\mathsf{Prov}_T(\ulcorner\bot\urcorner)$ *for some fixed contradiction $\bot$.*

*Proof (Sketch).* The key observation is that the argument of Case 1 in the First Incompleteness Theorem can itself be formalised inside $T$. That is, the reasoning

"if $T \vdash G$ then $T$ is inconsistent"

can be carried out within $T$ to yield:

$$T \vdash \mathsf{Con}(T) \to G. \tag{$**$}$$

This formalisation requires checking that the proof of Case 1 uses only facts about provability that are themselves provable in $T$ — this is the content of the *Hilbert-Bernays-Löb derivability conditions*, which PA satisfies.

Now suppose for contradiction that $T \vdash \mathsf{Con}(T)$. Then by ($**$) and Modus Ponens, $T \vdash G$. But this contradicts Case 1 of the First Incompleteness Theorem (which showed $T \nvdash G$ under the assumption that $T$ is consistent). Therefore $T \nvdash \mathsf{Con}(T)$. $\square$

**Remark:** The Second Incompleteness Theorem is the formal death of Hilbert's Programme, which sought to prove the consistency of mathematics within mathematics itself. No sufficiently strong, consistent, recursively axiomatisable theory can prove its own consistency.

## Relationship to Our Framework

The incompleteness theorems do not contradict the Completeness Theorem proved earlier. The distinction is:

- **Gödel Completeness** (proved above) concerns the proof system for *first-order logic*: every logically valid sentence ($\models \varphi$) is provable ($\vdash \varphi$). This is a statement about the *rules of inference.*
- **Gödel Incompleteness** concerns a specific *theory* $T$ (a set of non-logical axioms such as PA): there are sentences $\varphi$ that are true in the intended model but which $T \nvdash \varphi$.

More precisely, the incompleteness theorems assert that $T$ is not *semantically complete* in the sense that $T \not\models G$ is impossible — in fact $\mathcal{M} \models G$, but $G$ is not a logical consequence of $T$ alone, so $T \nvdash G$ by the Completeness Theorem (running it in the contrapositive). The following table summarises the relationship:

| **Result** | **Concerns** | **Statement** |
|---|---|---|
| Soundness | Proof system | $\Gamma \vdash \varphi \Rightarrow \Gamma \models \varphi$ |
| Completeness (Gödel) | Proof system | $\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi$ |
| 1st Incompleteness | Arithmetic theories | $\exists G : T \nvdash G$ and $T \nvdash \neg G$ |
| 2nd Incompleteness | Arithmetic theories | $T \nvdash \mathsf{Con}(T)$ |
| Compactness | Both | $\Gamma$ sat. $\iff$ fin. subsets sat. |